

Blockchain Simulation

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This project implements a discrete-event simulation of a simplified blockchain using the SimPy framework. The system simulates block mining, network propagation, transaction handling, difficulty adjustments, and coin issuance using modular Python classes and user-defined parameters.

Architecture Overview

There were 5 primary classes that contributed to the project.

Blockchain

This class contains the blocks, number of transactions, transaction pool, and other information that is used often with the rest of the simulation. It would also create blocks and add transactions to the current block. As well create the reward transaction for the winning miner.

Block

This class contained the transactions as well as the size of the block. The block size is created from the amount of transactions in it as well as the header. It also contains a timestamp, block id, and other information. These objects went in the blockchain to obtain

Node

This class kept a ledger of the blocks from the blockchain. It also contacted other nodes and pushed the information to them to ensure that the other nodes had the same ledger.

Miner

The Miner object was used to mine blocks. Each miner is associated with a specific node. The miner is able to mine the block by obtaining a mining time based on the difficulty and hashrate. This value is $\exp(\text{hashrate} / \text{difficulty})$. This allowed for a poisson distribution of the mining times to achieve the desired time.

Note - In hindsight, this could have been implemented by inheriting from the Node class. If this is ever developed further, it should be refactored accordingly.

Wallet

This object is used to hold balances, track the amount of transactions in and out of it, as well as have specific IDs. And in the case of users, such as miners, is attached to them specifically.

Additional Classes

Stats - This class was not a part of the core logic completely. It was used by the primary mining loop to store specific variables such as difficulty and was used to create, calculate, and print the stats.

Project Flow

There are two main processes occurring throughout the program. The mining process and the transaction process. These two processes are both run using the Simpy package. Each process uses the `env.process()` function with timeouts for specific actions.

Additionally, object initialization occurs in a file called `init_objs.py`. This file has a few functions that initialize the miners, nodes, and wallets. This occurs before either process is started. As well as creating the Blockchain class which initializes the first block.

Mining Process

The mining process mostly takes place in the `begin_mining()` function. This function creates the stats class as well as initializes the difficulty. It then goes into a loop to mine blocks until a stop signal is created.

The process of mining blocks is having each miner mine the block, which they do by obtaining a time from $\exp(\text{total_hashrate} / \text{difficulty})$. The smallest time to mine is then used with the `simpy env process` for that timeout amount. The block is then processed and the next block is created. As well the winner is awarded the reward at the beginning of the transaction pool. This repeats

until the total blocks are reached either from the input 'years' or 'blocks' or the number of transactions have all been processed. These give a signal and end the process.

As well every 'print' the stats are displayed.

Transaction Process

The transactions process is the other process that runs concurrently with the mining process. It starts a loop that checks every 'interval' which is timed out in the simpy env. Each iteration, it runs through each wallet and if the balance is greater than 0, it gives 1/number of to another wallet. This is decided by calculating the wallets that were given the minimum transactions by the current wallet. This is an attempt to try and equalize the transactions among the wallets. This will be explained under [transactions issues](#).

These transactions are then added to the transaction pool which is only fully processed once the block is finalized. Except, the senders would receive the negative balance in their wallet immediately. This is to prevent transactions with coins that don't exist.

The stop signal was sent to the blockchain object if the number of transactions was reached. This would then stop the simulation after the transactions were processed in the final block

Implementation Issues

There were two major issues that were faced when implementing the blockchain simulation. These contributed to significant delays in producing the final product as they were either very difficult to solve or only appeared late in the development process.

Transactions issues

The reason for this potentially complex way to conduct transactions is to ensure each wallet can create enough transactions. The problem is that with especially low values of 'interval', too many transactions would be created in a specific amount of time from wallets since the only available wallets for transactions were the wallets associated with miners.

For example, using an interval of .01, 1000 transactions and 1000 wallets. Consider the first wallet received a reward of 50 and we want to make transactions at 1% of the wallet size. With any blocktime greater than 10 seconds in this example, the wallet would be reduced to $.99^{1000} = .00004317124 * 50$ or .002158. So any other wallet with this would then have only .002158 coins. But since these transactions are not processed until the block is complete, these wallets do not create any transactions until it is complete.

Then when these wallets have a balance greater than 0 to complete a transaction, they only have a miniscule amount of coins to do it. This then reduces their balance even more and if the transactions are random then. Therefore the implementation used equal amounts of coins given to each wallet. This helps reduce this issue.

Additionally it is to be noted that this would not be an issue in the real world implementation as there would typically be a minimum transaction size or a flat fee that would prevent this from occurring. The simulation would be able to handle the quantity and frequency of the transactions quite easily if there were more coins available in the economy, which is how it would be expected to be implemented. This is purely an issue because it would be difficult to do these transactions as soon as a blockchain is created with very little coins.

Memory Issues

Initially, the code created many instances of the block and transaction objects as these objects would be stored in the blockchain. Although this is how it would be in the real world, except implemented in the simulation it leads to large amounts of memory demand. This was not a problem for a majority of the simulations, but for Memo at 10 years it created memory problems.

To solve this problem, there was garbage collection implemented after 1 million blocks to limit the size of the stored blocks to just the previous 500,000. Blocks besides the previous block were not really used in the simulation and would be only for historical purposes, similarly to real world implementations of blockchains.

This led to no other issues after there were a few other adjustments with the code.

Runs and Analysis

The screenshots of each of the runs for both the 10 years of each coin and different transaction workloads of each coin are shown.

10 Year Runs

Bitcoin 10 years:


```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ time python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 --blocktime 600 --blocksize 4000 --wallets 10 --transactions 0 --interval 10.0 --print 30000 --reward 50 --halving 210000 --years 10

Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 600.0 | Print: 30000 | Transactions: 0 | Blocksize: 4000 | Interval: 10.0 | Reward: 50.0 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:30000/525600 5.71% ABT:598.66s Diff:31.138M H:10000 InFl:0% ETA:296696843s Tx:29998 TPS:0.0 C:1499.95K NMB:36.62 Pool:1 IO:59998
B:60000/525600 11.42% ABT:603.46s Diff:30.306M H:10000 InFl:3.48% ETA:280969804s Tx:59998 TPS:0.0 C:2999.95K NMB:73.24 Pool:1 IO:19998
B:90000/525600 17.12% ABT:598.91s Diff:30.777M H:10000 InFl:2.63% ETA:260883691s Tx:89998 TPS:0.0 C:4499.95K NMB:109.86 Pool:1 IO:17998
B:120000/525600 22.83% ABT:599.9s Diff:29.58M H:10000 InFl:2.34% ETA:243318120s Tx:119998 TPS:0.0 C:5999.95K NMB:146.48 Pool:1 IO:23998
B:150000/525600 28.54% ABT:599.95s Diff:29.643M H:10000 InFl:2.19% ETA:225341387s Tx:149998 TPS:0.0 C:7499.95K NMB:183.1 Pool:1 IO:29998
B:180000/525600 34.25% ABT:599.53s Diff:31.519M H:10000 InFl:2.1% ETA:207197925s Tx:179998 TPS:0.0 C:8999.95K NMB:219.73 Pool:1 IO:35998
B:210000/525600 39.95% ABT:602.34s Diff:29.457M H:10000 InFl:2.04% ETA:190098152s Tx:209998 TPS:0.0 C:10499.95K NMB:256.35 Pool:1 IO:41998
B:240000/525600 45.66% ABT:597.56s Diff:31.404M H:10000 InFl:1.88% ETA:170662721s Tx:239998 TPS:0.0 C:11249.95K NMB:292.97 Pool:1 IO:47998
B:270000/525600 51.37% ABT:602.96s Diff:30.588M H:10000 InFl:1.86% ETA:154116374s Tx:269998 TPS:0.0 C:11999.95K NMB:329.59 Pool:1 IO:53998
B:300000/525600 57.08% ABT:599.67s Diff:30.838M H:10000 InFl:1.86% ETA:135286254s Tx:299998 TPS:0.0 C:12749.95K NMB:366.21 Pool:1 IO:59998
B:330000/525600 62.79% ABT:599.94s Diff:29.073M H:10000 InFl:1.86% ETA:117348487s Tx:329998 TPS:0.0 C:13499.95K NMB:402.83 Pool:1 IO:65998
B:360000/525600 68.49% ABT:600.32s Diff:29.86M H:10000 InFl:1.85% ETA:99412635s Tx:359998 TPS:0.0 C:14249.95K NMB:439.45 Pool:1 IO:71998
B:390000/525600 74.2% ABT:599.84s Diff:29.53M H:10000 InFl:1.84% ETA:81338266s Tx:389998 TPS:0.0 C:14999.95K NMB:476.07 Pool:1 IO:77998
B:420000/525600 79.91% ABT:599.92s Diff:30.476M H:10000 InFl:1.84% ETA:63351069s Tx:419998 TPS:0.0 C:15749.95K NMB:512.69 Pool:1 IO:83998
B:450000/525600 85.62% ABT:600.9s Diff:29.532M H:10000 InFl:1.79% ETA:45428410s Tx:449998 TPS:0.0 C:16124.95K NMB:549.31 Pool:1 IO:89998
B:480000/525600 91.32% ABT:599.8s Diff:30.693M H:10000 InFl:1.79% ETA:27350771s Tx:479998 TPS:0.0 C:16499.95K NMB:585.94 Pool:1 IO:95998
B:510000/525600 97.03% ABT:599.78s Diff:29.018M H:10000 InFl:1.79% ETA:9356600s Tx:509998 TPS:0.0 C:16874.95K NMB:622.56 Pool:1 IO:101998
End: 8:525600/525600 100.0% ABT:601.98s Diff:29.809M H:10000 InFl:3.39% ETA:0s Tx:525598 TPS:0.0 C:17069.95K NMB:641.6 Pool:1 IO:1051198
python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 8.80s user 0.60s system 99% cpu 9.402 total
```

BCH 10 Years:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ time python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 --blocktime 600 --blocksize 128000 --wallets 10 --transactions 0 --interval 10.0 --print 30000 --reward 12.5 --halving 210000 --years 10

Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 600.0 | Print: 30000 | Transactions: 0 | Blocksize: 128000 | Interval: 10.0 | Reward: 12.5 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:30000/525600 5.71% ABT:600.67s Diff:29.411M H:10000 InFl:0% ETA:297692088s Tx:29998 TPS:0.0 C:374.99K NMB:36.62 Pool:1 IO:59998
B:60000/525600 11.42% ABT:598.93s Diff:31.269M H:10000 InFl:3.51% ETA:278860073s Tx:59998 TPS:0.0 C:749.99K NMB:73.24 Pool:1 IO:19998
B:90000/525600 17.12% ABT:599.75s Diff:31.475M H:10000 InFl:2.63% ETA:261251281s Tx:89998 TPS:0.0 C:1124.99K NMB:109.86 Pool:1 IO:17998
B:120000/525600 22.83% ABT:601.84s Diff:29.561M H:10000 InFl:2.33% ETA:244106246s Tx:119998 TPS:0.0 C:1499.99K NMB:146.48 Pool:1 IO:23998
B:150000/525600 28.54% ABT:599.26s Diff:30.829M H:10000 InFl:2.19% ETA:225080391s Tx:149998 TPS:0.0 C:1874.99K NMB:183.1 Pool:1 IO:29998
B:180000/525600 34.25% ABT:600.7s Diff:29.636M H:10000 InFl:2.1% ETA:207601262s Tx:179998 TPS:0.0 C:2249.99K NMB:219.73 Pool:1 IO:35998
B:210000/525600 39.95% ABT:600.95s Diff:29.329M H:10000 InFl:2.04% ETA:189661182s Tx:209998 TPS:0.0 C:2624.99K NMB:256.35 Pool:1 IO:41998
B:240000/525600 45.66% ABT:599.8s Diff:30.326M H:10000 InFl:1.88% ETA:171303013s Tx:239998 TPS:0.0 C:2812.49K NMB:292.97 Pool:1 IO:47998
B:270000/525600 51.37% ABT:601.4s Diff:29.442M H:10000 InFl:1.86% ETA:153717240s Tx:269998 TPS:0.0 C:2999.99K NMB:329.59 Pool:1 IO:53998
B:300000/525600 57.08% ABT:598.25s Diff:30.246M H:10000 InFl:1.87% ETA:134965565s Tx:299998 TPS:0.0 C:3187.49K NMB:366.21 Pool:1 IO:59998
B:330000/525600 62.79% ABT:602.55s Diff:30.045M H:10000 InFl:1.85% ETA:117858289s Tx:329998 TPS:0.0 C:3374.99K NMB:402.83 Pool:1 IO:65998
B:360000/525600 68.49% ABT:598.54s Diff:29.903M H:10000 InFl:1.85% ETA:99117548s Tx:359998 TPS:0.0 C:3562.49K NMB:439.45 Pool:1 IO:71998
B:390000/525600 74.2% ABT:601.44s Diff:29.565M H:10000 InFl:1.84% ETA:8154686s Tx:389998 TPS:0.0 C:3749.99K NMB:476.07 Pool:1 IO:77998
B:420000/525600 79.91% ABT:600.47s Diff:29.842M H:10000 InFl:1.84% ETA:63409143s Tx:419998 TPS:0.0 C:3937.49K NMB:512.69 Pool:1 IO:83998
B:450000/525600 85.62% ABT:600.52s Diff:29.538M H:10000 InFl:1.79% ETA:45399537s Tx:449998 TPS:0.0 C:4031.24K NMB:549.31 Pool:1 IO:89998
B:480000/525600 91.32% ABT:597.87s Diff:31.005M H:10000 InFl:1.8% ETA:27262724s Tx:479998 TPS:0.0 C:4124.99K NMB:585.94 Pool:1 IO:95998
B:510000/525600 97.03% ABT:601.64s Diff:29.785M H:10000 InFl:1.79% ETA:9385506s Tx:509998 TPS:0.0 C:4218.74K NMB:622.56 Pool:1 IO:101998
End: 8:525600/525600 100.0% ABT:600.37s Diff:31.327M H:10000 InFl:3.4% ETA:0s Tx:525598 TPS:0.0 C:4267.49K NMB:641.6 Pool:1 IO:1051198
python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 9.06s user 0.63s system 99% cpu 9.691 total
```

LTC 10 Years:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ time python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 --blocktime 150 --blocksize 4000 --wallets 10 --transactions 0 --interval 10.0 --print 100000 --reward 50 --halving 840000 --years 10

Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 150.0 | Print: 100000 | Transactions: 0 | Blocksize: 4000 | Interval: 10.0 | Reward: 50.0 | Halving: 840000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:100000/2102400 4.76% ABT:150.05s Diff:7.262M H:10000 InFl:0% ETA:300465655s Tx:99998 TPS:0.01 C:4999.95K NMB:122.07 Pool:1 IO:199998
B:200000/2102400 9.51% ABT:150.13s Diff:7.282M H:10000 InFl:4.2% ETA:285607683s Tx:199998 TPS:0.01 C:9999.95K NMB:244.14 Pool:1 IO:399998
B:300000/2102400 14.27% ABT:150.16s Diff:7.368M H:10000 InFl:3.15% ETA:270650216s Tx:299998 TPS:0.01 C:14999.95K NMB:366.21 Pool:1 IO:599998
B:400000/2102400 19.03% ABT:149.98s Diff:7.213M H:10000 InFl:2.8% ETA:255331161s Tx:399998 TPS:0.01 C:19999.95K NMB:488.28 Pool:1 IO:799998
B:500000/2102400 23.78% ABT:150.04s Diff:7.475M H:10000 InFl:2.63% ETA:240427352s Tx:499998 TPS:0.01 C:24999.95K NMB:610.35 Pool:1 IO:999998
B:600000/2102400 28.54% ABT:150.06s Diff:7.794M H:10000 InFl:2.52% ETA:225452148s Tx:599998 TPS:0.01 C:29999.95K NMB:732.42 Pool:1 IO:1199998
B:700000/2102400 33.3% ABT:150.2s Diff:7.344M H:10000 InFl:2.45% ETA:210641122s Tx:699998 TPS:0.01 C:34999.95K NMB:854.49 Pool:1 IO:1399998
B:800000/2102400 38.05% ABT:150.04s Diff:7.424M H:10000 InFl:2.4% ETA:195410573s Tx:799998 TPS:0.01 C:39999.95K NMB:976.56 Pool:1 IO:1599998
B:900000/2102400 42.81% ABT:150.03s Diff:7.458M H:10000 InFl:2.29% ETA:180401854s Tx:899998 TPS:0.01 C:43499.95K NMB:1098.63 Pool:1 IO:1799998
B:1000000/2102400 47.56% ABT:150.14s Diff:7.318M H:10000 InFl:2.22% ETA:165511451s Tx:999998 TPS:0.01 C:45999.95K NMB:1220.7 Pool:1 IO:1999998
B:1100000/2102400 52.32% ABT:149.98s Diff:7.491M H:10000 InFl:2.22% ETA:150335875s Tx:1099998 TPS:0.01 C:48499.95K NMB:1342.77 Pool:1 IO:2199998
B:1200000/2102400 57.08% ABT:150.06s Diff:7.524M H:10000 InFl:2.21% ETA:135410525s Tx:1199998 TPS:0.01 C:50999.95K NMB:1464.84 Pool:1 IO:2399998
B:1300000/2102400 61.83% ABT:150.07s Diff:7.54M H:10000 InFl:2.2% ETA:120414432s Tx:1299998 TPS:0.01 C:53499.95K NMB:1586.91 Pool:1 IO:2599998
B:1400000/2102400 66.59% ABT:150.03s Diff:7.448M H:10000 InFl:2.2% ETA:105300223s Tx:1399998 TPS:0.01 C:55999.95K NMB:1708.98 Pool:1 IO:2799998
B:1500000/2102400 71.35% ABT:150.18s Diff:7.271M H:10000 InFl:2.19% ETA:90467038s Tx:1499998 TPS:0.01 C:58499.95K NMB:1831.05 Pool:1 IO:2999998
B:1600000/2102400 76.1% ABT:149.9s Diff:7.645M H:10000 InFl:2.19% ETA:75310801s Tx:1599998 TPS:0.01 C:60999.95K NMB:1953.12 Pool:1 IO:3199998
B:1700000/2102400 80.86% ABT:150.08s Diff:7.567M H:10000 InFl:2.18% ETA:60393319s Tx:1699998 TPS:0.01 C:63249.95K NMB:2075.19 Pool:1 IO:3399998
B:1800000/2102400 85.62% ABT:150.13s Diff:7.683M H:10000 InFl:2.14% ETA:45398147s Tx:1799998 TPS:0.01 C:64499.95K NMB:2197.26 Pool:1 IO:3599998
B:1900000/2102400 90.37% ABT:150.02s Diff:7.705M H:10000 InFl:2.14% ETA:30364355s Tx:1899998 TPS:0.01 C:65749.95K NMB:2319.33 Pool:1 IO:3799998
B:2000000/2102400 95.13% ABT:150.13s Diff:7.586M H:10000 InFl:2.14% ETA:15373335s Tx:1999998 TPS:0.01 C:66999.95K NMB:2441.4 Pool:1 IO:3999998
B:2100000/2102400 99.89% ABT:150.1s Diff:7.462M H:10000 InFl:2.14% ETA:360245s Tx:2099998 TPS:0.01 C:68249.95K NMB:2563.48 Pool:1 IO:4199998
End: 8:2102400/2102400 100.0% ABT:150.16s Diff:7.491M H:10000 InFl:85.9% ETA:0s Tx:2102398 TPS:0.01 C:68279.95K NMB:2566.4 Pool:1 IO:4204798
python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 58.43s user 1.14s system 99% cpu 59.593 total
```

Dogecoin 10 years:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ time python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 --blocktime 60 --blocksize 4000 --wallets 10 --transactions 0 --interval 10.0 --print 300000 --reward 10000 --halving 0 --years 10
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 60.0 | Print: 300000 | Transactions: 0 | Blocksize: 4000 | Interval: 10.0 | Reward: 10000.0 | Halving: 0 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:300000/5256000 5.71% ABT:60.03s Diff:2.967M H:10000 Infl:0% ETA:297510820s Tx:299998 TPS:0.02 C:2999990.0K NMB:366.21 Pool:1 I0:599998
B:600000/5256000 11.42% ABT:60.02s Diff:2.972M H:10000 Infl:3.5% ETA:279472492s Tx:599998 TPS:0.02 C:5999990.0K NMB:732.42 Pool:1 I0:1199998
B:900000/5256000 17.12% ABT:60.03s Diff:3.101M H:10000 Infl:2.63% ETA:261488786s Tx:899998 TPS:0.02 C:8999990.0K NMB:1098.63 Pool:1 I0:1799998
B:1200000/5256000 22.83% ABT:60.03s Diff:3.003M H:10000 Infl:2.33% ETA:243482319s Tx:1199998 TPS:0.02 C:11999990.0K NMB:1464.84 Pool:1 I0:2399998
B:1500000/5256000 28.54% ABT:60.02s Diff:3.019M H:10000 Infl:2.19% ETA:225445843s Tx:1499998 TPS:0.02 C:14999990.0K NMB:1831.05 Pool:1 I0:2999998
B:1800000/5256000 34.25% ABT:60.05s Diff:3.036M H:10000 Infl:2.1% ETA:207518501s Tx:1799998 TPS:0.02 C:17999990.0K NMB:2197.26 Pool:1 I0:3599998
B:2100000/5256000 39.95% ABT:60.01s Diff:3.082M H:10000 Infl:2.04% ETA:189379054s Tx:2099998 TPS:0.02 C:20999990.0K NMB:2563.48 Pool:1 I0:4199998
B:2400000/5256000 45.66% ABT:60.02s Diff:3.104M H:10000 Infl:2.0% ETA:171430415s Tx:2399998 TPS:0.02 C:23999990.0K NMB:2929.69 Pool:1 I0:4799998
B:2700000/5256000 51.37% ABT:60.04s Diff:3.048M H:10000 Infl:1.97% ETA:153471589s Tx:2699998 TPS:0.02 C:26999990.0K NMB:3295.9 Pool:1 I0:5399998
B:3000000/5256000 57.08% ABT:60.01s Diff:3.122M H:10000 Infl:1.95% ETA:135391658s Tx:2999998 TPS:0.02 C:29999990.0K NMB:3662.11 Pool:1 I0:5999998
B:3300000/5256000 62.79% ABT:60.05s Diff:2.918M H:10000 Infl:1.93% ETA:117451967s Tx:3299998 TPS:0.02 C:32999990.0K NMB:4028.32 Pool:1 I0:6599998
B:3600000/5256000 68.49% ABT:60.02s Diff:3.015M H:10000 Infl:1.91% ETA:99396468s Tx:3599998 TPS:0.02 C:35999990.0K NMB:4394.53 Pool:1 I0:7199998
B:3900000/5256000 74.2% ABT:60.02s Diff:2.956M H:10000 Infl:1.9% ETA:81387337s Tx:3899998 TPS:0.02 C:38999990.0K NMB:4760.74 Pool:1 I0:7799998
B:4200000/5256000 79.91% ABT:60.05s Diff:3.021M H:10000 Infl:1.89% ETA:63407538s Tx:4199998 TPS:0.02 C:41999990.0K NMB:5126.95 Pool:1 I0:8399998
B:4500000/5256000 85.62% ABT:60.02s Diff:3.042M H:10000 Infl:1.88% ETA:45378169s Tx:4499998 TPS:0.02 C:44999990.0K NMB:5493.16 Pool:1 I0:8999998
B:4800000/5256000 91.32% ABT:60.03s Diff:3.004M H:10000 Infl:1.87% ETA:27374423s Tx:4799998 TPS:0.02 C:47999990.0K NMB:5859.37 Pool:1 I0:9599998
B:5100000/5256000 97.03% ABT:60.04s Diff:2.967M H:10000 Infl:1.86% ETA:9366638s Tx:5099998 TPS:0.02 C:50999990.0K NMB:6225.58 Pool:1 I0:10199998
End: B:5256000/5256000 100.0% ABT:60.03s Diff:3.037M H:10000 Infl:3.47% ETA:0s Tx:5255998 TPS:0.02 C:5255990.0K NMB:6416.01 Pool:1 I0:10511998
python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 179.54s user 1.47s system 99% cpu 3:01.07 total
```

Memo 10 Years:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 --blocktime 3.27 --blocksize 32000 --wallets 10 --transactions 0 --interval 10 --print 5000000 --reward 51.8457072 --halving 9644000 --years 10
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 3.27 | Print: 5000000 | Transactions: 0 | Blocksize: 32000 | Interval: 10.0 | Reward: 51.8457072 | Halving: 9644000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:5000000/96440367 5.18% ABT:3.27s Diff:0.16M H:10000 Infl:0% ETA:299160175s Tx:4999998 TPS:0.31 C:259228.48K NMB:6103.51 Pool:1 I0:9999998
B:10000000/96440367 10.37% ABT:3.27s Diff:0.165M H:10000 Infl:3.79% ETA:282801462s Tx:9999998 TPS:0.31 C:509228.48K NMB:12207.03 Pool:1 I0:19999998
B:15000000/96440367 15.55% ABT:3.27s Diff:0.157M H:10000 Infl:2.42% ETA:266446781s Tx:14999998 TPS:0.31 C:638842.75K NMB:18310.55 Pool:1 I0:29999998
B:20000000/96440367 20.74% ABT:3.27s Diff:0.159M H:10000 Infl:2.29% ETA:250085740s Tx:19999998 TPS:0.31 C:759228.48K NMB:24414.06 Pool:1 I0:39999998
B:25000000/96440367 25.92% ABT:3.27s Diff:0.168M H:10000 Infl:2.09% ETA:233718074s Tx:24999998 TPS:0.31 C:824035.62K NMB:30517.58 Pool:1 I0:49999998
B:30000000/96440367 31.11% ABT:3.27s Diff:0.163M H:10000 Infl:2.06% ETA:217374785s Tx:29999998 TPS:0.31 C:881921.35K NMB:36621.09 Pool:1 I0:59999998
B:35000000/96440367 36.29% ABT:3.27s Diff:0.164M H:10000 Infl:2.0% ETA:201007083s Tx:34999998 TPS:0.31 C:914324.92K NMB:42724.61 Pool:1 I0:69999998
B:40000000/96440367 41.48% ABT:3.27s Diff:0.165M H:10000 Infl:1.99% ETA:184655474s Tx:39999998 TPS:0.31 C:942114.22K NMB:48828.12 Pool:1 I0:79999998
B:45000000/96440367 46.66% ABT:3.27s Diff:0.162M H:10000 Infl:1.96% ETA:168292290s Tx:44999998 TPS:0.31 C:958316.0K NMB:54931.64 Pool:1 I0:89999998
B:50000000/96440367 51.85% ABT:3.27s Diff:0.168M H:10000 Infl:1.95% ETA:151934191s Tx:49999998 TPS:0.31 C:971633.87K NMB:61035.15 Pool:1 I0:99999998
B:55000000/96440367 57.03% ABT:3.27s Diff:0.164M H:10000 Infl:1.94% ETA:135582073s Tx:54999998 TPS:0.31 C:979734.76K NMB:67138.67 Pool:1 I0:109999998
B:60000000/96440367 62.21% ABT:3.27s Diff:0.159M H:10000 Infl:1.94% ETA:119220016s Tx:59999998 TPS:0.31 C:986105.3K NMB:73242.19 Pool:1 I0:119999998
B:65000000/96440367 67.4% ABT:3.27s Diff:0.16M H:10000 Infl:1.94% ETA:102857985s Tx:64999998 TPS:0.31 C:990155.75K NMB:79345.7 Pool:1 I0:129999998
B:70000000/96440367 72.58% ABT:3.27s Diff:0.162M H:10000 Infl:1.93% ETA:86503367s Tx:69999998 TPS:0.31 C:993196.82K NMB:85449.22 Pool:1 I0:139999998
B:75000000/96440367 77.77% ABT:3.27s Diff:0.167M H:10000 Infl:1.93% ETA:70144330s Tx:74999998 TPS:0.31 C:995222.04K NMB:91552.73 Pool:1 I0:149999998
B:80000000/96440367 82.95% ABT:3.27s Diff:0.176M H:10000 Infl:1.93% ETA:53789131s Tx:79999998 TPS:0.31 C:996670.48K NMB:97656.25 Pool:1 I0:159999998
B:85000000/96440367 88.14% ABT:3.27s Diff:0.161M H:10000 Infl:1.93% ETA:37428659s Tx:84999998 TPS:0.31 C:997683.09K NMB:103759.76 Pool:1 I0:169999998
B:90000000/96440367 93.32% ABT:3.27s Diff:0.16M H:10000 Infl:1.93% ETA:21071061s Tx:89999998 TPS:0.31 C:998371.26K NMB:109863.28 Pool:1 I0:179999998
B:95000000/96440367 98.51% ABT:3.27s Diff:0.162M H:10000 Infl:1.93% ETA:4712380s Tx:94999998 TPS:0.31 C:998877.57K NMB:115966.8 Pool:1 I0:189999998
End: B:96440367/96440367 100.0% ABT:3.27s Diff:0.162M H:10000 Infl:6.69% ETA:0s Tx:96440365 TPS:0.31 C:999023.4K NMB:117725.06 Pool:1 I0:192888732
```

	Infl after 10y	Coins After 10y	Time	Blocks
BTC	1.79%	17.1m	8.8s	525,600
BCH	1.79%	4.3m	9.06s	525,600
LTC	2.14%	68.3m	58.43s	2,102,400
DogeCoin	1.86%	52.6b	179.54s	5,256,000
Memo	1.93%	5t	~100 mins	96,440,367

*Memo was not run with the traditional time command.

Notes

It seems that the most important aspect of the time taken for each command was dependent on the amount of blocks. And it seems to be relatively linear with perhaps a few more time spent as the block number gets larger. As well, it's interesting that after 10 years the lack of a halving

from dogecoin hardly affected its difference in inflation compared to a coin like bitcoin. This is the same case as well for memo as the halving only occurred towards the end. This makes sense with their size of coins after 10 years.

The blocks seem to have similar amounts of coins after 10 years as the true coins in the real world which is a testament to the accuracy of the simulation.

Transaction Runs

BTC Small:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (main) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1 --blocktime 600 --blocksize 4000 --wallets 10 --transactions 10 --interval 10.0 --print 1 --reward 50 --halving 210000 --years 10
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 600.0 | Print: 1 | Transactions: 10 | Blocksize: 4000 | Interval: 10.0 | Reward: 50.0 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:2/525600 0.0% ABT:584.1s Diff:30.0M H:10000 Infl:0% ETA:307002352s Tx:0 TPS:0.0 C:0.05K NMB:0.0 Pool:1 IO:2
B:3/525600 0.0% ABT:1318.63s Diff:30.0M H:10000 Infl:47831.46% ETA:693067948s Tx:1 TPS:0.0 C:0.1K NMB:0.0 Pool:1 IO:4
B:4/525600 0.0% ABT:2848.03s Diff:30.0M H:10000 Infl:16609.39% ETA:1496912135s Tx:12 TPS:0.0 C:0.15K NMB:0.01 Pool:1 IO:6
B:5/525600 0.0% ABT:307.51s Diff:30.0M H:10000 Infl:136739.02% ETA:161623350s Tx:23 TPS:0.04 C:0.2K NMB:0.01 Pool:1 IO:8
B:6/525600 0.0% ABT:354.55s Diff:30.0M H:10000 Infl:111183.48% ETA:186348869s Tx:44 TPS:0.06 C:0.25K NMB:0.02 Pool:1 IO:10
B:7/525600 0.0% ABT:727.77s Diff:30.0M H:10000 Infl:51998.8% ETA:382511139s Tx:55 TPS:0.02 C:0.3K NMB:0.02 Pool:1 IO:12
B:8/525600 0.0% ABT:271.46s Diff:30.0M H:10000 Infl:135533.89% ETA:142677089s Tx:66 TPS:0.04 C:0.35K NMB:0.02 Pool:1 IO:14
B:9/525600 0.0% ABT:280.53s Diff:30.0M H:10000 Infl:128476.9% ETA:147442072s Tx:87 TPS:0.07 C:0.4K NMB:0.03 Pool:1 IO:16
B:10/525600 0.0% ABT:174.93s Diff:30.0M H:10000 Infl:202813.59% ETA:91940988s Tx:98 TPS:0.06 C:0.45K NMB:0.03 Pool:1 IO:18
End: B:11/525600 0.0% ABT:1183.72s Diff:30.0M H:10000 Infl:29601.53% ETA:622151522s Tx:109 TPS:0.01 C:0.5K NMB:0.04 Pool:1 IO:20
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (main) $
```

BTC Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1 --blocktime 600 --blocksize 4000 --wallets 1000 --transactions 1000 --interval 1 --print 10 --reward 50 --halving 210000 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 600.0 | Print: 10 | Transactions: 1000 | Blocksize: 4000 | Interval: 1.0 | Reward: 50.0 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:10/525600 0.0% ABT:194.91s Diff:90.0M H:10000 Infl:0% ETA:102443822s Tx:7462 TPS:3.83 C:0.45K NMB:1.83 Pool:1 IO:18
B:20/525600 0.0% ABT:1057.82s Diff:90.0M H:10000 Infl:6293.72% ETA:555967371s Tx:47472 TPS:3.78 C:0.95K NMB:11.61 Pool:66269 IO:38
B:30/525600 0.01% ABT:787.01s Diff:90.0M H:10000 Infl:6116.05% ETA:413628647s Tx:87482 TPS:5.08 C:1.45K NMB:21.39 Pool:132102 IO:58
B:40/525600 0.01% ABT:885.36s Diff:90.0M H:10000 Infl:4790.2% ETA:465309352s Tx:127492 TPS:4.52 C:1.95K NMB:31.16 Pool:224897 IO:78
B:50/525600 0.01% ABT:902.58s Diff:90.0M H:10000 Infl:4389.85% ETA:474353330s Tx:167502 TPS:4.43 C:2.45K NMB:40.94 Pool:327361 IO:98
B:60/525600 0.01% ABT:205.63s Diff:90.0M H:10000 Infl:18466.04% ETA:108067354s Tx:207512 TPS:19.46 C:2.95K NMB:50.72 Pool:320273 IO:118
B:70/525600 0.01% ABT:472.76s Diff:90.0M H:10000 Infl:7801.29% ETA:248447452s Tx:247522 TPS:8.46 C:3.45K NMB:60.5 Pool:359001 IO:138
B:80/525600 0.02% ABT:461.39s Diff:90.0M H:10000 Infl:7825.66% ETA:242467270s Tx:287532 TPS:8.67 C:3.95K NMB:70.28 Pool:397439 IO:158
B:90/525600 0.02% ABT:511.02s Diff:90.0M H:10000 Infl:6952.37% ETA:268545191s Tx:327542 TPS:7.83 C:4.45K NMB:80.05 Pool:444309 IO:178
B:100/525600 0.02% ABT:560.73s Diff:90.0M H:10000 Infl:6256.04% ETA:294662485s Tx:367552 TPS:7.14 C:4.95K NMB:89.83 Pool:499645 IO:198
B:110/525600 0.02% ABT:315.15s Diff:90.0M H:10000 Infl:11017.34% ETA:165609680s Tx:407562 TPS:12.7 C:5.45K NMB:99.61 Pool:513212 IO:218
B:120/525600 0.02% ABT:481.63s Diff:90.0M H:10000 Infl:7148.41% ETA:253089461s Tx:447572 TPS:8.31 C:5.95K NMB:109.39 Pool:552551 IO:238
B:130/525600 0.02% ABT:607.56s Diff:90.0M H:10000 Infl:5626.82% ETA:319252569s Tx:487582 TPS:6.59 C:6.45K NMB:119.16 Pool:512551 IO:258
B:140/525600 0.03% ABT:621.89s Diff:90.0M H:10000 Infl:5464.05% ETA:326780799s Tx:527592 TPS:6.43 C:6.95K NMB:128.94 Pool:472551 IO:278
B:150/525600 0.03% ABT:457.55s Diff:90.0M H:10000 Infl:7388.25% ETA:240418561s Tx:567602 TPS:8.74 C:7.45K NMB:138.72 Pool:432551 IO:298
B:160/525600 0.03% ABT:951.66s Diff:90.0M H:10000 Infl:3536.2% ETA:500038735s Tx:607612 TPS:4.2 C:7.95K NMB:148.5 Pool:392551 IO:318
B:170/525600 0.03% ABT:716.59s Diff:90.0M H:10000 Infl:4677.63% ETA:376517452s Tx:647622 TPS:5.58 C:8.45K NMB:158.28 Pool:352551 IO:338
B:180/525600 0.03% ABT:669.84s Diff:90.0M H:10000 Infl:4986.58% ETA:351946576s Tx:687632 TPS:5.97 C:8.95K NMB:168.05 Pool:312551 IO:358
B:190/525600 0.04% ABT:526.69s Diff:90.0M H:10000 Infl:6322.07% ETA:276729006s Tx:727642 TPS:7.6 C:9.45K NMB:177.83 Pool:272551 IO:378
B:200/525600 0.04% ABT:561.21s Diff:90.0M H:10000 Infl:5916.59% ETA:294860412s Tx:767652 TPS:7.13 C:9.95K NMB:187.61 Pool:232551 IO:398
B:210/525600 0.04% ABT:274.49s Diff:90.0M H:10000 Infl:12066.32% ETA:144213789s Tx:807662 TPS:14.58 C:10.45K NMB:197.39 Pool:192551 IO:418
B:220/525600 0.04% ABT:447.63s Diff:90.0M H:10000 Infl:7382.24% ETA:235174378s Tx:847672 TPS:8.94 C:10.95K NMB:207.17 Pool:152551 IO:438
B:230/525600 0.04% ABT:819.52s Diff:90.0M H:10000 Infl:4023.79% ETA:430553807s Tx:887682 TPS:4.88 C:11.45K NMB:216.94 Pool:112551 IO:458
B:240/525600 0.05% ABT:450.22s Diff:90.0M H:10000 Infl:7310.49% ETA:236526361s Tx:927692 TPS:8.89 C:11.95K NMB:226.72 Pool:72551 IO:478
B:250/525600 0.05% ABT:439.99s Diff:90.0M H:10000 Infl:7467.36% ETA:231147670s Tx:967702 TPS:9.09 C:12.45K NMB:236.5 Pool:32551 IO:498
End: B:259/525600 0.05% ABT:339.36s Diff:90.0M H:10000 Infl:10247.37% ETA:178278594s Tx:1000261 TPS:10.21 C:12.9K NMB:244.46 Pool:1 IO:516
```

BTC Large:


```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 600 --blocksize 4000 --wallets 1000 --transactions 1000 --interval .01 --print 10 --reward 210000 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 600.0 | Print: 10 | Transactions: 1000 | Blocksize: 4000 | Interval: 0.01
| Reward: 50.0 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:10/525600 0.0% ABT:495.78s Diff:90.0M H:10000 Infl:0% ETA:260575865s Tx:28008 TPS:5.65 C:0.45K NMB:6.85 Pool:972005 IO:18
B:20/525600 0.0% ABT:862.32s Diff:90.0M H:10000 Infl:7720.53% ETA:4532204709s Tx:68018 TPS:4.64 C:0.95K NMB:16.62 Pool:932005 IO:38
B:30/525600 0.01% ABT:1061.04s Diff:90.0M H:10000 Infl:4536.48% ETA:557651082s Tx:108028 TPS:3.77 C:1.45K NMB:26.4 Pool:892005 IO:58
B:40/525600 0.01% ABT:472.75s Diff:90.0M H:10000 Infl:8970.97% ETA:248459784s Tx:148038 TPS:8.46 C:1.95K NMB:36.18 Pool:852005 IO:78
B:50/525600 0.01% ABT:403.55s Diff:90.0M H:10000 Infl:9818.35% ETA:212086847s Tx:188048 TPS:9.91 C:2.45K NMB:45.96 Pool:812005 IO:98
B:60/525600 0.01% ABT:613.11s Diff:90.0M H:10000 Infl:6193.31% ETA:322214998s Tx:228058 TPS:6.53 C:2.95K NMB:55.74 Pool:772005 IO:118
B:70/525600 0.01% ABT:772.55s Diff:90.0M H:10000 Infl:4773.93% ETA:405998823s Tx:268068 TPS:5.18 C:3.45K NMB:65.51 Pool:732005 IO:138
B:80/525600 0.02% ABT:572.62s Diff:90.0M H:10000 Infl:6305.53% ETA:380920744s Tx:308078 TPS:6.99 C:3.95K NMB:75.29 Pool:692005 IO:158
B:90/525600 0.02% ABT:457.93s Diff:90.0M H:10000 Infl:7758.35% ETA:240647479s Tx:348088 TPS:8.74 C:4.45K NMB:85.07 Pool:652005 IO:178
B:100/525600 0.02% ABT:361.69s Diff:90.0M H:10000 Infl:9698.61% ETA:190070650s Tx:388098 TPS:11.06 C:4.95K NMB:94.85 Pool:612005 IO:198
B:110/525600 0.02% ABT:1130.46s Diff:90.0M H:10000 Infl:3071.45% ETA:594044391s Tx:428108 TPS:3.54 C:5.45K NMB:104.62 Pool:572005 IO:218
B:120/525600 0.02% ABT:916.16s Diff:90.0M H:10000 Infl:3757.98% ETA:481425365s Tx:468118 TPS:4.37 C:5.95K NMB:114.4 Pool:532005 IO:238
B:130/525600 0.02% ABT:576.78s Diff:90.0M H:10000 Infl:5927.01% ETA:303082863s Tx:508128 TPS:6.94 C:6.45K NMB:124.18 Pool:492005 IO:258
B:140/525600 0.03% ABT:539.69s Diff:90.0M H:10000 Infl:6296.33% ETA:283585434s Tx:548138 TPS:7.41 C:6.95K NMB:133.96 Pool:452005 IO:278
B:150/525600 0.03% ABT:575.37s Diff:90.0M H:10000 Infl:5875.27% ETA:302330507s Tx:588148 TPS:6.95 C:7.45K NMB:143.74 Pool:412005 IO:298
B:160/525600 0.03% ABT:521.95s Diff:90.0M H:10000 Infl:6447.48% ETA:274252359s Tx:628158 TPS:7.67 C:7.95K NMB:153.51 Pool:372005 IO:318
B:170/525600 0.03% ABT:705.77s Diff:90.0M H:10000 Infl:4749.34% ETA:370832268s Tx:668168 TPS:5.67 C:8.45K NMB:163.29 Pool:332005 IO:338
B:180/525600 0.03% ABT:447.33s Diff:90.0M H:10000 Infl:7467.02% ETA:323503467s Tx:708178 TPS:8.94 C:8.95K NMB:173.07 Pool:292005 IO:358
B:190/525600 0.04% ABT:428.95s Diff:90.0M H:10000 Infl:7762.65% ETA:225374058s Tx:748188 TPS:9.33 C:9.45K NMB:182.85 Pool:252005 IO:378
B:200/525600 0.04% ABT:921.74s Diff:90.0M H:10000 Infl:3602.36% ETA:484284573s Tx:788198 TPS:4.34 C:9.95K NMB:192.63 Pool:212005 IO:398
B:210/525600 0.04% ABT:654.13s Diff:90.0M H:10000 Infl:5063.36% ETA:343670825s Tx:828208 TPS:6.12 C:10.45K NMB:202.4 Pool:172005 IO:418
B:220/525600 0.04% ABT:345.02s Diff:90.0M H:10000 Infl:9577.74% ETA:181265331s Tx:868218 TPS:11.6 C:10.95K NMB:212.18 Pool:132005 IO:438
B:230/525600 0.04% ABT:854.16s Diff:90.0M H:10000 Infl:3860.64% ETA:448749365s Tx:908228 TPS:4.68 C:11.45K NMB:221.96 Pool:92005 IO:458
B:240/525600 0.05% ABT:747.62s Diff:90.0M H:10000 Infl:4402.36% ETA:392772215s Tx:948238 TPS:5.35 C:11.95K NMB:231.74 Pool:52005 IO:478
B:250/525600 0.05% ABT:561.87s Diff:90.0M H:10000 Infl:5847.55% ETA:295177119s Tx:988248 TPS:7.12 C:12.45K NMB:241.51 Pool:12005 IO:498
End: B:254/525600 0.05% ABT:651.77s Diff:90.0M H:10000 Infl:14970.19% ETA:342404043s Tx:1000256 TPS:5.61 C:12.65K NMB:244.45 Pool:1 IO:506
```

BCH Small:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 600 --blocksize 128000 --wallets 10 --transactions 10 --interval 10 --print 10 --reward 12.5 --halving 210000 --years 10
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 600.0 | Print: 10 | Transactions: 10 | Blocksize: 128000 | Interval: 10.0 | R
eward: 12.5 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
End: B:5/525600 0.0% ABT:344.13s Diff:30.0M H:10000 Infl:0% ETA:180871528s Tx:105 TPS:0.03 C:0.05K NMB:0.03 Pool:1 IO:8
```

BCH Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 600 --blocksize 128000 --wallets 1000 --transactions 1000 --interval 1 --print 10 --reward 12.5 --halving 210000 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 600.0 | Print: 10 | Transactions: 1000 | Blocksize: 128000 | Interval: 1.0
| Reward: 12.5 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:10/525600 0.0% ABT:600.55s Diff:90.0M H:10000 Infl:0% ETA:315640824s Tx:19537 TPS:3.25 C:0.11K NMB:4.78 Pool:1 IO:18
B:20/525600 0.0% ABT:491.21s Diff:90.0M H:10000 Infl:13553.48% ETA:258169968s Tx:75114 TPS:11.31 C:0.24K NMB:18.36 Pool:1 IO:38
B:30/525600 0.01% ABT:1001.65s Diff:90.0M H:10000 Infl:4805.47% ETA:526436377s Tx:213648 TPS:13.83 C:0.36K NMB:52.19 Pool:1 IO:58
B:40/525600 0.01% ABT:501.4s Diff:90.0M H:10000 Infl:8458.4% ETA:263516150s Tx:290767 TPS:15.38 C:0.49K NMB:71.03 Pool:1 IO:78
B:50/525600 0.01% ABT:366.31s Diff:90.0M H:10000 Infl:10816.59% ETA:192513818s Tx:353048 TPS:17.0 C:0.61K NMB:86.24 Pool:1 IO:98
B:60/525600 0.01% ABT:599.19s Diff:90.0M H:10000 Infl:6337.16% ETA:314900467s Tx:454922 TPS:17.0 C:0.74K NMB:111.12 Pool:1 IO:118
B:70/525600 0.01% ABT:695.18s Diff:90.0M H:10000 Infl:5305.24% ETA:365339229s Tx:573099 TPS:17.0 C:0.86K NMB:139.98 Pool:1 IO:138
B:80/525600 0.02% ABT:428.35s Diff:90.0M H:10000 Infl:8429.16% ETA:225107198s Tx:645937 TPS:17.0 C:0.99K NMB:157.78 Pool:1 IO:158
B:90/525600 0.02% ABT:527.07s Diff:90.0M H:10000 Infl:6740.6% ETA:276982258s Tx:735554 TPS:17.0 C:1.11K NMB:179.67 Pool:1 IO:178
B:100/525600 0.02% ABT:713.59s Diff:90.0M H:10000 Infl:4915.91% ETA:374990894s Tx:856876 TPS:17.0 C:1.24K NMB:209.29 Pool:1 IO:198
B:110/525600 0.02% ABT:799.69s Diff:90.0M H:10000 Infl:4341.88% ETA:420227905s Tx:992818 TPS:17.0 C:1.36K NMB:242.49 Pool:1 IO:218
End: B:112/525600 0.02% ABT:628.69s Diff:90.0M H:10000 Infl:39042.31% ETA:330369464s Tx:1000113 TPS:8.87 C:1.39K NMB:244.28 Pool:1 IO:222
```

BCH Large:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 600 --blocksize 128000 --wallets 1000 --transactions 1000 --interval .01 --print 10 --reward 12.5 --halving 210000 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 600.0 | Print: 10 | Transactions: 1000 | Blocksize: 128000 | Interval: 0.0
1 | Reward: 12.5 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:10/525600 0.0% ABT:578.98s Diff:90.0M H:10000 Infl:0% ETA:304306788s Tx:850795 TPS:146.95 C:0.11K NMB:207.72 Pool:149215 IO:18
End: B:12/525600 0.0% ABT:625.12s Diff:90.0M H:10000 Infl:51620.68% ETA:328555599s Tx:1000011 TPS:199.84 C:0.14K NMB:244.15 Pool:1 IO:22
```

LTC Small:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 150 --blocksize 4000 --wallets 10 --transactions 10 --interval 10 --print 2 --reward 50 --halving 840000 --years 10
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 150.0 | Print: 2 | Transactions: 10 | Blocksize: 4000 | Interval: 10.0 | Rewa
rd: 50.0 | Halving: 840000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:2/2102400 0.0% ABT:11.56s Diff:7.5M H:10000 Infl:0% ETA:24293941s Tx:0 TPS:0.0 C:0.05K NMB:0.0 Pool:1 IO:2
B:4/2102400 0.0% ABT:74.53s Diff:7.5M H:10000 Infl:634708.19% ETA:156688920s Tx:10 TPS:0.07 C:0.15K NMB:0.01 Pool:1 IO:6
End: B:6/2102400 0.0% ABT:145.55s Diff:7.5M H:10000 Infl:180551.08% ETA:306012652s Tx:104 TPS:0.32 C:0.25K NMB:0.03 Pool:1 IO:10
```

LTC Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 150 --blocksize 4000 --wallets 1000 --transactions 1000 --interval .1 --print 25 --reward 50 --halving 210000 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 150.0 | Print: 25 | Transactions: 1000 | Blocksize: 4000 | Interval: 1.0 | Rewa
rd: 50.0 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:25/2102400 0.0% ABT:149.68s Diff:22.5M H:10000 Infl:0% ETA:314686175s Tx:28923 TPS:17.73 C:1.2K NMB:7.08 Pool:4178 IO:48
B:50/2102400 0.0% ABT:127.71s Diff:22.5M H:10000 Infl:20166.96% ETA:268482463s Tx:83572 TPS:17.12 C:2.45K NMB:20.45 Pool:1 IO:98
B:75/2102400 0.0% ABT:134.93s Diff:22.5M H:10000 Infl:14119.04% ETA:283659191s Tx:140938 TPS:17.01 C:3.7K NMB:34.48 Pool:1 IO:148
B:100/2102400 0.0% ABT:153.35s Diff:22.5M H:10000 Infl:11004.69% ETA:323294099s Tx:286141 TPS:17.01 C:4.95K NMB:50.42 Pool:1 IO:198
B:125/2102400 0.01% ABT:232.31s Diff:22.5M H:10000 Infl:6801.33% ETA:488369723s Tx:291966 TPS:14.78 C:6.2K NMB:71.4 Pool:12937 IO:248
B:150/2102400 0.01% ABT:147.4s Diff:22.5M H:10000 Infl:10283.12% ETA:389878009s Tx:366499 TPS:20.23 C:7.45K NMB:89.62 Pool:1074 IO:298
B:175/2102400 0.01% ABT:139.2s Diff:22.5M H:10000 Infl:10582.5% ETA:292631099s Tx:426757 TPS:17.32 C:8.7K NMB:104.36 Pool:1 IO:348
B:200/2102400 0.01% ABT:208.5s Diff:22.5M H:10000 Infl:6919.39% ETA:438305153s Tx:515386 TPS:17.0 C:9.95K NMB:126.02 Pool:1 IO:398
B:225/2102400 0.01% ABT:158.78s Diff:22.5M H:10000 Infl:8942.72% ETA:333780458s Tx:582901 TPS:17.01 C:11.2K NMB:142.53 Pool:1 IO:448
B:250/2102400 0.01% ABT:106.24s Diff:22.5M H:10000 Infl:13199.27% ETA:223322142s Tx:627539 TPS:16.81 C:12.45K NMB:153.45 Pool:540 IO:498
B:275/2102400 0.01% ABT:81.62s Diff:22.5M H:10000 Infl:17005.78% ETA:171585163s Tx:662783 TPS:17.27 C:13.7K NMB:162.08 Pool:1 IO:548
B:300/2102400 0.01% ABT:114.5s Diff:22.5M H:10000 Infl:12021.66% ETA:240700055s Tx:711479 TPS:17.01 C:14.95K NMB:173.99 Pool:1 IO:598
B:325/2102400 0.02% ABT:172.54s Diff:22.5M H:10000 Infl:7922.24% ETA:362694357s Tx:784286 TPS:16.88 C:16.2K NMB:191.79 Pool:540 IO:648
B:350/2102400 0.02% ABT:183.51s Diff:22.5M H:10000 Infl:7404.48% ETA:385740668s Tx:858191 TPS:16.11 C:17.45K NMB:209.86 Pool:4656 IO:698
B:375/2102400 0.02% ABT:190.55s Diff:22.5M H:10000 Infl:7094.06% ETA:400549099s Tx:939900 TPS:17.15 C:18.7K NMB:229.83 Pool:3960 IO:748
End: B:398/2102400 0.02% ABT:136.99s Diff:22.5M H:10000 Infl:10039.14% ETA:287948718s Tx:1000407 TPS:18.15 C:19.85K NMB:244.63 Pool:1 IO:794
```

LTC Large:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 150 --blocksize 4000 --wallets 1000 --transactions 1000 --interval .01 --print 10 --reward 50 --halving 840000 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 150.0 | Print: 10 | Transactions: 1000 | Blocksize: 4000 | Interval: 0.01
| Reward: 50.0 | Halving: 840000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:10/2102400 0.0% ABT:151.67s Diff:22.5M H:10000 Infl:0% ETA:318878842s Tx:26768 TPS:17.65 C:0.45K NMB:6.54 Pool:825074 IO:18
B:20/2102400 0.0% ABT:180.75s Diff:22.5M H:10000 Infl:36834.17% ETA:379995102s Tx:66778 TPS:22.14 C:0.95K NMB:16.32 Pool:933244 IO:38
B:30/2102400 0.0% ABT:215.21s Diff:22.5M H:10000 Infl:22366.39% ETA:452443448s Tx:106788 TPS:18.59 C:1.45K NMB:26.1 Pool:893244 IO:58
B:40/2102400 0.0% ABT:119.64s Diff:22.5M H:10000 Infl:35449.0% ETA:251522225s Tx:146798 TPS:33.44 C:1.95K NMB:35.88 Pool:853244 IO:78
B:50/2102400 0.0% ABT:110.09s Diff:22.5M H:10000 Infl:35989.39% ETA:231456080s Tx:186808 TPS:36.34 C:2.45K NMB:45.66 Pool:813244 IO:98
B:60/2102400 0.0% ABT:162.97s Diff:22.5M H:10000 Infl:23299.48% ETA:342625182s Tx:226818 TPS:24.55 C:2.95K NMB:55.43 Pool:773244 IO:118
B:70/2102400 0.0% ABT:161.18s Diff:22.5M H:10000 Infl:22882.16% ETA:338850109s Tx:266828 TPS:24.82 C:3.45K NMB:65.21 Pool:733244 IO:138
B:80/2102400 0.0% ABT:158.5s Diff:22.5M H:10000 Infl:22780.46% ETA:333212205s Tx:306838 TPS:25.24 C:3.95K NMB:74.99 Pool:693244 IO:158
B:90/2102400 0.0% ABT:276.62s Diff:22.5M H:10000 Infl:12843.61% ETA:581539494s Tx:346848 TPS:14.46 C:4.45K NMB:84.77 Pool:653244 IO:178
B:100/2102400 0.0% ABT:158.76s Diff:22.5M H:10000 Infl:22096.13% ETA:333756875s Tx:386858 TPS:25.2 C:4.95K NMB:94.54 Pool:613244 IO:198
B:110/2102400 0.01% ABT:234.02s Diff:22.5M H:10000 Infl:14837.25% ETA:491968245s Tx:426868 TPS:17.1 C:5.45K NMB:104.32 Pool:573244 IO:218
B:120/2102400 0.01% ABT:211.53s Diff:22.5M H:10000 Infl:16276.11% ETA:444699999s Tx:466878 TPS:18.91 C:5.95K NMB:114.1 Pool:533244 IO:238
B:130/2102400 0.01% ABT:140.97s Diff:22.5M H:10000 Infl:24250.17% ETA:296362330s Tx:506888 TPS:28.38 C:6.45K NMB:123.88 Pool:493244 IO:258
B:140/2102400 0.01% ABT:184.68s Diff:22.5M H:10000 Infl:18399.87% ETA:388242766s Tx:546898 TPS:21.66 C:6.95K NMB:133.66 Pool:453244 IO:278
B:150/2102400 0.01% ABT:106.28s Diff:22.5M H:10000 Infl:31806.97% ETA:223429313s Tx:586908 TPS:37.65 C:7.45K NMB:143.43 Pool:413244 IO:298
B:160/2102400 0.01% ABT:94.46s Diff:22.5M H:10000 Infl:35624.68% ETA:198586052s Tx:626918 TPS:42.35 C:7.95K NMB:153.21 Pool:373244 IO:318
B:170/2102400 0.01% ABT:175.87s Diff:22.5M H:10000 Infl:19059.2% ETA:369718891s Tx:666928 TPS:22.75 C:8.45K NMB:162.99 Pool:333244 IO:338
B:180/2102400 0.01% ABT:92.6s Diff:22.5M H:10000 Infl:36069.44% ETA:194675695s Tx:706938 TPS:43.21 C:8.95K NMB:172.77 Pool:293244 IO:358
B:190/2102400 0.01% ABT:89.11s Diff:22.5M H:10000 Infl:37033.11% ETA:189017178s Tx:746948 TPS:44.5 C:9.45K NMB:182.54 Pool:253244 IO:378
B:200/2102400 0.01% ABT:220.11s Diff:22.5M H:10000 Infl:15085.74% ETA:467066264s Tx:786958 TPS:18.18 C:9.95K NMB:192.32 Pool:213244 IO:398
B:210/2102400 0.01% ABT:116.66s Diff:22.5M H:10000 Infl:28392.0% ETA:245231268s Tx:826698 TPS:34.3 C:10.45K NMB:202.1 Pool:173244 IO:418
B:220/2102400 0.01% ABT:236.64s Diff:22.5M H:10000 Infl:13964.25% ETA:497458472s Tx:866978 TPS:16.91 C:10.95K NMB:211.88 Pool:133244 IO:438
B:230/2102400 0.01% ABT:122.69s Diff:22.5M H:10000 Infl:26878.24% ETA:257980899s Tx:906988 TPS:32.61 C:11.45K NMB:221.66 Pool:93244 IO:458
B:240/2102400 0.01% ABT:121.99s Diff:22.5M H:10000 Infl:26979.79% ETA:256446197s Tx:946998 TPS:32.8 C:11.95K NMB:231.43 Pool:53244 IO:478
B:250/2102400 0.01% ABT:171.43s Diff:22.5M H:10000 Infl:19165.9% ETA:360364938s Tx:987008 TPS:23.34 C:12.45K NMB:241.21 Pool:13244 IO:498
End: B:254/2102400 0.01% ABT:102.74s Diff:22.5M H:10000 Infl:74509.35% ETA:215971920s Tx:1000255 TPS:30.8 C:12.65K NMB:244.45 Pool:1 IO:506
```

Dogecoin Small:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 60 --blocksize 4000 --wallets 10 --transactions 10 --interval 10 --print 10 --reward 10 --halving 0 --years 10
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 60.0 | Print: 10 | Transactions: 10 | Blocksize: 4000 | Interval: 10.0 | Rewa
rd: 10.0 | Halving: 0 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:10/5256000 0.0% ABT:23.01s Diff:3.0M H:10000 Infl:0% ETA:120927213s Tx:79 TPS:0.34 C:0.09K NMB:0.03 Pool:1 IO:18
End: B:12/5256000 0.0% ABT:32.05s Diff:3.0M H:10000 Infl:355644.35% ETA:168454912s Tx:110 TPS:0.29 C:0.11K NMB:0.04 Pool:1 IO:22
```

Dogecoin Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 60 --blocksize 4000 --wallets 1000 --transactions 1000 --interval 1 --print 100 --reward 10 --halving 0 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 60.0 | Print: 100 | Transactions: 1000 | Blocksize: 4000 | Interval: 1.0 | Rewa
rd: 10.0 | Halving: 0 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:100/5256000 0.0% ABT:59.37s Diff:9.0M H:10000 Infl:0% ETA:312036226s Tx:79904 TPS:13.46 C:0.99K NMB:19.6 Pool:1 IO:198
B:200/5256000 0.0% ABT:65.8s Diff:9.0M H:10000 Infl:9634.27% ETA:345815413s Tx:101864 TPS:17.02 C:1.99K NMB:47.04 Pool:1 IO:398
B:300/5256000 0.01% ABT:69.0s Diff:9.0M H:10000 Infl:6866.86% ETA:362658019s Tx:109264 TPS:17.01 C:2.99K NMB:75.8 Pool:1 IO:598
B:400/5256000 0.01% ABT:61.81s Diff:9.0M H:10000 Infl:6808.32% ETA:324855819s Tx:414441 TPS:17.02 C:3.99K NMB:101.57 Pool:1 IO:798
B:500/5256000 0.01% ABT:57.12s Diff:9.0M H:10000 Infl:6904.93% ETA:300184847s Tx:511645 TPS:17.02 C:4.99K NMB:125.4 Pool:1 IO:998
B:600/5256000 0.01% ABT:55.57s Diff:9.0M H:10000 Infl:6812.33% ETA:292040489s Tx:606214 TPS:17.02 C:5.99K NMB:148.59 Pool:1 IO:1198
B:700/5256000 0.01% ABT:58.99s Diff:9.0M H:10000 Infl:6238.79% ETA:309094564s Tx:706597 TPS:17.02 C:6.99K NMB:173.19 Pool:1 IO:1398
B:800/5256000 0.02% ABT:60.31s Diff:9.0M H:10000 Infl:5977.52% ETA:316916332s Tx:800207 TPS:17.02 C:7.99K NMB:198.34 Pool:1 IO:1598
B:900/5256000 0.02% ABT:53.91s Diff:9.0M H:10000 Infl:6592.13% ETA:283291840s Tx:900954 TPS:17.02 C:8.99K NMB:220.84 Pool:1 IO:1798
B:1000/5256000 0.02% ABT:55.17s Diff:9.0M H:10000 Infl:6351.45% ETA:289942889s Tx:994860 TPS:17.02 C:9.99K NMB:243.86 Pool:1 IO:1998
End: B:1008/5256000 0.02% ABT:53.23s Diff:9.0M H:10000 Infl:777384.38% ETA:279737847s Tx:1001022 TPS:15.0 C:10.07K NMB:245.37 Pool:1 IO:2014
```

Dogecoin Large:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 60 --blocksize 4000 --wallets 1000 --transactions 1000 --interval .01 --print 100 --reward 10 --halving 0 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 60.0 | Print: 100 | Transactions: 1000 | Blocksize: 4000 | Interval: 0.01
| Reward: 10.0 | Halving: 0 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:100/5256000 0.0% ABT:64.66s Diff:9.0M H:10000 Infl:0% ETA:339842036s Tx:388098 TPS:60.02 C:0.99K NMB:94.85 Pool:612005 IO:198
B:200/5256000 0.0% ABT:62.18s Diff:9.0M H:10000 Infl:10195.33% ETA:326784965s Tx:788198 TPS:64.35 C:1.99K NMB:192.63 Pool:212005 IO:398
End: B:254/5256000 0.0% ABT:50.46s Diff:9.0M H:10000 Infl:14412.85% ETA:265216864s Tx:1000256 TPS:76.23 C:2.53K NMB:244.45 Pool:1 IO:506
```

Memo Small:


```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/S025/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 3.27 --blocksize 32000 --wallets 10 --transactions 10 --interval 10 --print 10 --reward 51.8457072 --halving 964400 --years 10
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 3.27 | Print: 10 | Transactions: 10 | Blocksize: 32000 | Interval: 10.0 | Rew
ard: 51.8457072 | Halving: 964400 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:10/96440367 0.0% ABT:2.73s Diff:0.164M H:10000 Infl:0% ETA:262844265s Tx:15 TPS:0.55 C:0.47K NMB:0.01 Pool:1 IO:18
B:20/96440367 0.0% ABT:3.12s Diff:0.164M H:10000 Infl:2137187.36% ETA:300423475s Tx:43 TPS:0.9 C:0.99K NMB:0.03 Pool:1 IO:38
B:30/96440367 0.0% ABT:3.95s Diff:0.164M H:10000 Infl:1218791.73% ETA:380873032s Tx:77 TPS:0.86 C:1.5K NMB:0.05 Pool:1 IO:58
B:40/96440367 0.0% ABT:2.59s Diff:0.164M H:10000 Infl:1640564.62% ETA:249309340s Tx:105 TPS:1.08 C:2.02K NMB:0.06 Pool:1 IO:78
B:50/96440367 0.0% ABT:3.05s Diff:0.164M H:10000 Infl:1299813.36% ETA:293978597s Tx:133 TPS:0.92 C:2.54K NMB:0.08 Pool:1 IO:98
End: B:58/96440367 0.0% ABT:4.56s Diff:0.164M H:10000 Infl:930344.86% ETA:440232060s Tx:159 TPS:0.66 C:2.96K NMB:0.09 Pool:1 IO:114
```

Memo Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/S025/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 3.27 --blocksize 32000 --wallets 1000 --transactions 1000 --interval 1 --print 1000 --reward 51.8457072 --halving 964400 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 3.27 | Print: 1000 | Transactions: 1000 | Blocksize: 32000 | Interval: 1.0
| Reward: 51.8457072 | Halving: 964400 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:1000/96440367 0.0% ABT:3.4s Diff:0.49M H:10000 Infl:0% ETA:327804087s Tx:57877 TPS:17.03 C:51.79K NMB:15.11 Pool:1 IO:1998
B:2000/96440367 0.0% ABT:3.34s Diff:0.49M H:10000 Infl:18879.28% ETA:322343063s Tx:115691 TPS:17.3 C:103.64K NMB:30.2 Pool:1 IO:3998
B:3000/96440367 0.0% ABT:3.1s Diff:0.475M H:10000 Infl:15261.51% ETA:298963463s Tx:169391 TPS:17.32 C:155.49K NMB:44.28 Pool:1 IO:5998
B:4000/96440367 0.0% ABT:3.16s Diff:0.475M H:10000 Infl:13286.68% ETA:305214923s Tx:224196 TPS:17.32 C:207.33K NMB:58.64 Pool:1 IO:7998
B:5000/96440367 0.01% ABT:3.15s Diff:0.498M H:10000 Infl:12525.01% ETA:303526408s Tx:278712 TPS:17.32 C:259.18K NMB:72.93 Pool:1 IO:9998
B:6000/96440367 0.01% ABT:3.3s Diff:0.498M H:10000 Infl:11472.85% ETA:318099388s Tx:335778 TPS:17.3 C:311.02K NMB:87.84 Pool:1 IO:11998
B:7000/96440367 0.01% ABT:3.25s Diff:0.503M H:10000 Infl:11306.96% ETA:313794552s Tx:392096 TPS:17.31 C:362.87K NMB:102.56 Pool:1 IO:13998
B:8000/96440367 0.01% ABT:3.28s Diff:0.503M H:10000 Infl:10981.69% ETA:316489847s Tx:448890 TPS:17.3 C:414.71K NMB:117.4 Pool:1 IO:15998
B:9000/96440367 0.01% ABT:3.46s Diff:0.505M H:10000 Infl:10243.61% ETA:333987758s Tx:508778 TPS:17.29 C:466.56K NMB:133.0 Pool:1 IO:17998
B:10000/96440367 0.01% ABT:3.34s Diff:0.505M H:10000 Infl:10478.19% ETA:322475240s Tx:566626 TPS:17.3 C:518.41K NMB:148.1 Pool:1 IO:19998
B:11000/96440367 0.01% ABT:3.25s Diff:0.484M H:10000 Infl:10675.64% ETA:313342126s Tx:622859 TPS:17.31 C:570.25K NMB:162.81 Pool:1 IO:21998
B:12000/96440367 0.01% ABT:3.26s Diff:0.484M H:10000 Infl:10548.39% ETA:314497510s Tx:679313 TPS:17.31 C:622.1K NMB:177.57 Pool:1 IO:23998
B:13000/96440367 0.01% ABT:3.23s Diff:0.486M H:10000 Infl:10582.0% ETA:311317733s Tx:735189 TPS:17.31 C:673.94K NMB:192.18 Pool:1 IO:25998
B:14000/96440367 0.01% ABT:3.41s Diff:0.486M H:10000 Infl:9944.96% ETA:329295889s Tx:794244 TPS:17.29 C:725.79K NMB:207.58 Pool:1 IO:27998
B:15000/96440367 0.02% ABT:3.38s Diff:0.475M H:10000 Infl:10007.46% ETA:325566262s Tx:852653 TPS:17.3 C:777.63K NMB:222.81 Pool:1 IO:29998
B:16000/96440367 0.02% ABT:3.15s Diff:0.475M H:10000 Infl:10663.42% ETA:304177731s Tx:907271 TPS:17.31 C:829.48K NMB:237.13 Pool:1 IO:31998
B:17000/96440367 0.02% ABT:3.33s Diff:0.48M H:10000 Infl:10053.44% ETA:321369499s Tx:964932 TPS:17.3 C:881.33K NMB:252.18 Pool:1 IO:33998
End: B:17985/96440367 0.02% ABT:3.12s Diff:0.48M H:10000 Infl:10884.65% ETA:300890375s Tx:1017988 TPS:17.31 C:932.39K NMB:266.09 Pool:1 IO:35968
```

Memo Large:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/S025/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 3.27 --blocksize 32000 --wallets 1000 --transactions 1000 --interval .01 --print 10 --reward 51.8457072 --halving 964400 --years 10
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 3.27 | Print: 10 | Transactions: 1000 | Blocksize: 32000 | Interval: 0.01
| Reward: 51.8457072 | Halving: 964400 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0
B:10/96440367 0.0% ABT:4.07s Diff:0.49M H:10000 Infl:0% ETA:392403242s Tx:14191 TPS:348.77 C:0.47K NMB:3.47 Pool:1 IO:18
B:20/96440367 0.0% ABT:3.51s Diff:0.49M H:10000 Infl:1898504.29% ETA:338193206s Tx:50621 TPS:1038.85 C:0.99K NMB:12.38 Pool:1 IO:38
B:30/96440367 0.0% ABT:3.93s Diff:0.49M H:10000 Infl:1225797.0% ETA:378696394s Tx:101844 TPS:1304.47 C:1.5K NMB:24.89 Pool:1 IO:58
B:40/96440367 0.0% ABT:2.92s Diff:0.49M H:10000 Infl:1452236.43% ETA:281640147s Tx:148280 TPS:1590.08 C:2.02K NMB:36.24 Pool:1 IO:78
B:50/96440367 0.0% ABT:3.85s Diff:0.49M H:10000 Infl:1029376.28% ETA:371212464s Tx:209874 TPS:1600.2 C:2.54K NMB:51.29 Pool:1 IO:98
B:60/96440367 0.0% ABT:1.91s Diff:0.49M H:10000 Infl:1988916.59% ETA:184121520s Tx:240493 TPS:1603.78 C:3.06K NMB:58.77 Pool:1 IO:118
B:70/96440367 0.0% ABT:2.28s Diff:0.49M H:10000 Infl:1617379.05% ETA:219912751s Tx:279263 TPS:1700.22 C:3.58K NMB:68.25 Pool:1 IO:138
B:80/96440367 0.0% ABT:3.85s Diff:0.49M H:10000 Infl:936809.15% ETA:371667760s Tx:344791 TPS:1700.32 C:4.1K NMB:84.25 Pool:1 IO:158
B:90/96440367 0.0% ABT:1.9s Diff:0.49M H:10000 Infl:1867107.29% ETA:183509561s Tx:377152 TPS:1700.68 C:4.61K NMB:92.17 Pool:1 IO:178
B:100/96440367 0.0% ABT:2.21s Diff:0.49M H:10000 Infl:1584149.13% ETA:213557150s Tx:414800 TPS:1700.15 C:5.13K NMB:101.37 Pool:1 IO:198
B:110/96440367 0.0% ABT:2.98s Diff:0.49M H:10000 Infl:1163972.56% ETA:287682555s Tx:465521 TPS:1700.33 C:5.65K NMB:113.76 Pool:1 IO:218
B:120/96440367 0.0% ABT:3.09s Diff:0.49M H:10000 Infl:1114759.11% ETA:297854629s Tx:518027 TPS:1700.05 C:6.17K NMB:126.59 Pool:1 IO:238
B:130/96440367 0.0% ABT:3.36s Diff:0.49M H:10000 Infl:1018623.55% ETA:254597851s Tx:575106 TPS:1700.75 C:6.69K NMB:140.53 Pool:1 IO:258
B:140/96440367 0.0% ABT:2.05s Diff:0.49M H:10000 Infl:1657066.61% ETA:197765237s Tx:609966 TPS:1699.95 C:7.21K NMB:149.05 Pool:1 IO:278
B:150/96440367 0.0% ABT:2.62s Diff:0.49M H:10000 Infl:1288271.41% ETA:253063135s Tx:654584 TPS:1700.35 C:7.73K NMB:159.96 Pool:1 IO:298
B:160/96440367 0.0% ABT:3.02s Diff:0.49M H:10000 Infl:1113125.19% ETA:291562445s Tx:705985 TPS:1700.19 C:8.24K NMB:172.51 Pool:1 IO:318
B:170/96440367 0.0% ABT:2.64s Diff:0.49M H:10000 Infl:1269695.38% ETA:254597851s Tx:750875 TPS:1700.41 C:8.76K NMB:183.48 Pool:1 IO:338
B:180/96440367 0.0% ABT:4.4s Diff:0.49M H:10000 Infl:814019.43% ETA:395727478s Tx:820653 TPS:1700.51 C:9.28K NMB:200.53 Pool:1 IO:358
B:190/96440367 0.0% ABT:2.83s Diff:0.49M H:10000 Infl:1174891.3% ETA:273322693s Tx:868841 TPS:1700.28 C:9.8K NMB:212.3 Pool:1 IO:378
B:200/96440367 0.0% ABT:2.5s Diff:0.49M H:10000 Infl:1330810.21% ETA:240624425s Tx:911266 TPS:1700.36 C:10.32K NMB:222.67 Pool:1 IO:398
B:210/96440367 0.0% ABT:3.23s Diff:0.49M H:10000 Infl:1025969.4% ETA:311331682s Tx:966152 TPS:1700.18 C:10.84K NMB:236.08 Pool:1 IO:418
End: B:212/96440367 0.0% ABT:5.25s Diff:0.49M H:10000 Infl:1274725.35% ETA:506010869s Tx:1000222 TPS:1364.1 C:10.94K NMB:244.4 Pool:1 IO:422
```

Table of some of the Interesting stats:

Coin	Workload	Blocks	Simulation Time (min)	Largest Pool	Max TPS
BTC	Small	11	110	0	0.07
BTC	Medium	259	2590	512k	19.46
BTC	Large	254	2540	972k	11.06
BCH	Small	5	50	0	0.03
BCH	Medium	112	1120	0	17

BCH	Large	12	120	150k	200
LTC	Small	6	15	0	0.32
LTC	Medium	398	995	13k	20
LTC	Large	254	635	933k	45
DogeCoin	Small	12	12	0	0.29
DogeCoin	Medium	1008	1008	0	17
DogeCoin	Large	254	254	612k	76
Memo	Small	58	3.161	0	1.08
Memo	Medium	17985	980.1825	0	17
Memo	Large	212	11.554	0	1700

Notes

Some interesting stats from the gathered simulation is quite interesting. The simulation time for Large transactions would take less than the medium transactions. This can be explained by the frequency of the transactions. In an ideal scenario the large runs would be completed in 10 seconds compared to the medium transaction which would take 1000 seconds. As well, the fastest transactions per second was Memo during the large run. This again makes sense by the fact that its block sizes are bigger than the others besides BCH and the blocks have the shortest time.

Latency, Bandwidth, and Fees

Bandwidth

Bandwidth is input as bytes/second, the command below has a bandwidth of 1MB/s.

```
(lit) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 100 --blocksize 4000 --wallets 10 --transactions 0 --interval 10 --print 1000 --reward 50 --halving 0 --years 10 --bandwidth 1048576 --blocks 1000
0
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 100.0 | Print: 1000 | Transactions: 0 | Blocksize: 4000 | Interval: 10.0 | Re
ward: 50.0 | Halving: 0 | Years: 10 | Blocks: 10000 | Difficulty: None | Latency: 0 | Bandwidth: 1048576 | Fee: 0
B:1000/10000 10.0% ABT:98.78s Diff:5.0M H:10000 Infl:0% ETA:889054s Tx:998 TPS:0.01 C:49.95K NMB:1.22 Pool:1 IO:1998 Network Time:2.44s
B:2000/10000 20.0% ABT:96.48s Diff:5.0M H:10000 Infl:654.08% ETA:771830s Tx:1998 TPS:0.01 C:99.95K NMB:2.44 Pool:1 IO:3998 Network Time:2.44s
B:3000/10000 30.0% ABT:102.8s Diff:5.122M H:10000 Infl:460.22% ETA:719632s Tx:2998 TPS:0.01 C:149.95K NMB:3.66 Pool:1 IO:5998 Network Time:2.44s
B:4000/10000 40.0% ABT:101.58s Diff:5.122M H:10000 Infl:413.99% ETA:609474s Tx:3998 TPS:0.01 C:199.95K NMB:4.88 Pool:1 IO:7998 Network Time:2.44s
B:5000/10000 50.0% ABT:97.45s Diff:5.015M H:10000 Infl:404.53% ETA:487262s Tx:4998 TPS:0.01 C:249.95K NMB:6.1 Pool:1 IO:9998 Network Time:2.44s
B:6000/10000 60.0% ABT:95.51s Diff:5.015M H:10000 Infl:396.24% ETA:382046s Tx:5998 TPS:0.01 C:299.95K NMB:7.32 Pool:1 IO:11998 Network Time:2.44s
B:7000/10000 70.0% ABT:104.4s Diff:5.195M H:10000 Infl:352.42% ETA:313205s Tx:6998 TPS:0.01 C:349.95K NMB:8.54 Pool:1 IO:13998 Network Time:2.44s
B:8000/10000 80.0% ABT:100.24s Diff:5.195M H:10000 Infl:359.55% ETA:280489s Tx:7998 TPS:0.01 C:399.95K NMB:9.76 Pool:1 IO:15998 Network Time:2.44s
B:9000/10000 90.0% ABT:100.21s Diff:5.076M H:10000 Infl:354.07% ETA:100205s Tx:8998 TPS:0.01 C:449.95K NMB:10.98 Pool:1 IO:17998 Network Time:2.44s
End: B:10000/10000 100.0% ABT:106.32s Diff:5.076M H:10000 Infl:329.58% ETA:0s Tx:9998 TPS:0.01 C:499.95K NMB:12.21 Pool:1 IO:19998 Network Time:2.44s
Avg Broadcast Time per block: 0.001220556640625
Total Broadcast Time: 24.4111328125
```

Broadcast time can be shown to be correct by multiplying the NMB by the number of nodes(2).

Latency

Latency is input as seconds, the command below has a latency of 100ms or .1s.

```
(iit) jacob@jacob-hp-ubuntu ~/Documents/iit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 100 --blocksize 4000 --wallets 10 --transactions 0 --interval 10 --print 1000 --reward 50 --halving 0 --years 10 --latency .1 --blocks 10000
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 100.0 | Print: 1000 | Transactions: 0 | Blocksize: 4000 | Interval: 10.0 | Re
ward: 50.0 | Halving: 0 | Years: 10 | Blocks: 10000 | Difficulty: None | Latency: 0.1 | Bandwidth: inf | Fee: 0
B:1000/10000 10.0% ABT:102.07s Diff:5.0M H:10000 Infl:0% ETA:918659s Tx:998 TPS:0.01 C:49.95K NMB:1.22 Pool:1 IO:1998 Network Time:99.9s
B:2000/10000 20.0% ABT:104.97s Diff:5.0M H:10000 Infl:601.75% ETA:839728s Tx:1998 TPS:0.01 C:99.95K NMB:2.44 Pool:1 IO:3998 Network Time:100.0s
B:3000/10000 30.0% ABT:95.69s Diff:4.833M H:10000 Infl:494.95% ETA:669818s Tx:2998 TPS:0.01 C:149.95K NMB:3.66 Pool:1 IO:5998 Network Time:100.0s
B:4000/10000 40.0% ABT:97.94s Diff:4.833M H:10000 Infl:429.8% ETA:587638s Tx:3998 TPS:0.01 C:199.95K NMB:4.88 Pool:1 IO:7998 Network Time:100.0s
B:5000/10000 50.0% ABT:93.34s Diff:4.969M H:10000 Infl:422.78% ETA:466722s Tx:4998 TPS:0.01 C:249.95K NMB:6.1 Pool:1 IO:9998 Network Time:100.0s
B:6000/10000 60.0% ABT:105.28s Diff:4.969M H:10000 Infl:359.81% ETA:421112s Tx:5998 TPS:0.01 C:299.95K NMB:7.32 Pool:1 IO:11998 Network Time:100.0s
B:7000/10000 70.0% ABT:100.0s Diff:5.025M H:10000 Infl:368.29% ETA:300008s Tx:6998 TPS:0.01 C:349.95K NMB:8.54 Pool:1 IO:13998 Network Time:100.0s
B:8000/10000 80.0% ABT:94.79s Diff:5.025M H:10000 Infl:380.63% ETA:189579s Tx:7998 TPS:0.01 C:399.95K NMB:9.76 Pool:1 IO:15998 Network Time:100.0s
B:9000/10000 90.0% ABT:104.81s Diff:5.119M H:10000 Infl:338.84% ETA:104805s Tx:8998 TPS:0.01 C:449.95K NMB:10.98 Pool:1 IO:17998 Network Time:100.0s
End: B:10000/10000 100.0% ABT:105.14s Diff:5.119M H:10000 Infl:333.59% ETA:0s Tx:9998 TPS:0.01 C:499.95K NMB:12.21 Pool:1 IO:19998 Network Time:100.0s
Avg Broadcast Time per block: 0.04999500000000000000
Total Broadcast Time: 999.9000000000000000
```

Latency can be shown to be correct by dividing the number of IO requests by the number of nodes(2) to obtain the total broadcast time.

Latency and Bandwidth

Both can be used together which affects the total network time. Network time is the cumulative sum of the time spent by latency and bandwidth.

```
(iit) jacob@jacob-hp-ubuntu ~/Documents/iit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 100 --blocksize 4000 --wallets 10 --transactions 0 --interval 10 --print 1000 --reward 50 --halving 0 --years 10 --bandwidth 1048576 --latency .1
--blocks 10000
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 100.0 | Print: 1000 | Transactions: 0 | Blocksize: 4000 | Interval: 10.0 | Re
ward: 50.0 | Halving: 0 | Years: 10 | Blocks: 10000 | Difficulty: None | Latency: 0.1 | Bandwidth: 1048576 | Fee: 0
B:1000/10000 10.0% ABT:99.88s Diff:5.0M H:10000 Infl:0% ETA:898917s Tx:998 TPS:0.01 C:49.95K NMB:1.22 Pool:1 IO:1998 Network Time:102.34s
B:2000/10000 20.0% ABT:97.14s Diff:5.0M H:10000 Infl:650.3% ETA:777122s Tx:1998 TPS:0.01 C:99.95K NMB:2.44 Pool:1 IO:3998 Network Time:102.44s
B:3000/10000 30.0% ABT:101.43s Diff:5.072M H:10000 Infl:466.94% ETA:709981s Tx:2998 TPS:0.01 C:149.95K NMB:3.66 Pool:1 IO:5998 Network Time:102.44s
B:4000/10000 40.0% ABT:103.05s Diff:5.072M H:10000 Infl:408.46% ETA:618315s Tx:3998 TPS:0.01 C:199.95K NMB:4.88 Pool:1 IO:7998 Network Time:102.44s
B:5000/10000 50.0% ABT:102.31s Diff:4.989M H:10000 Infl:385.72% ETA:511536s Tx:4998 TPS:0.01 C:249.95K NMB:6.1 Pool:1 IO:9998 Network Time:102.44s
B:6000/10000 60.0% ABT:96.46s Diff:4.989M H:10000 Infl:392.75% ETA:385843s Tx:5998 TPS:0.01 C:299.95K NMB:7.32 Pool:1 IO:11998 Network Time:102.44s
B:7000/10000 70.0% ABT:97.55s Diff:4.989M H:10000 Infl:377.58% ETA:292639s Tx:6998 TPS:0.01 C:349.95K NMB:8.54 Pool:1 IO:13998 Network Time:102.44s
B:8000/10000 80.0% ABT:98.97s Diff:4.989M H:10000 Infl:364.55% ETA:197935s Tx:7998 TPS:0.01 C:399.95K NMB:9.76 Pool:1 IO:15998 Network Time:102.44s
B:9000/10000 90.0% ABT:100.53s Diff:5.087M H:10000 Infl:353.29% ETA:100526s Tx:8998 TPS:0.01 C:449.95K NMB:10.98 Pool:1 IO:17998 Network Time:102.44s
End: B:10000/10000 100.0% ABT:98.64s Diff:5.087M H:10000 Infl:355.59% ETA:0s Tx:9998 TPS:0.01 C:499.95K NMB:12.21 Pool:1 IO:19998 Network Time:102.44s
Avg Broadcast Time per block: 0.05121555664062501
Total Broadcast Time: 1024.3111328125
```

The network time display is the aggregated time by print statement of the extra time taken. The average broadcast time per block allows for the user to see the total time cost with bandwidth and latency. As well the total broadcast time is the total time in the simulation that was limited by bandwidth and latency.

Currently latency and bandwidth are at currently reasonable levels at 1MB/s and 100ms latency. This hardly affects the blocktime. But if bandwidth were to decrease or latency increase this can be seen to clearly affect the blocktime:


```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 100 --blocksize 4000 --wallets 10 --transactions 0 --interval 10 --print 1000 --reward 50 --halving 0 --years 10 --bandwidth 256 --latency 1 --blo
cks 10000
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 10 | Hashrate: 10000 | Blocktime: 100.0 | Print: 1000 | Transactions: 0 | Blocksize: 4000 | Interval: 10.0 | Re
ward: 50.0 | Halving: 0 | Years: 10 | Blocks: 10000 | Difficulty: None | Latency: 1.0 | Bandwidth: 256 | Fee: 0
B:1000/10000 10.0% ABT:122.35s Diff:5.0M H:10000 Infl:0% ETA:1101118s Tx:998 TPS:0.01 C:49.95K NMB:1.22 Pool:1 IO:1998 Network Time:10987.0s
B:2000/10000 20.0% ABT:114.06s Diff:5.0M H:10000 Infl:612.29% ETA:912489s Tx:1998 TPS:0.01 C:99.95K NMB:2.44 Pool:1 IO:3998 Network Time:11000.0s
B:3000/10000 30.0% ABT:100.25s Diff:4.217M H:10000 Infl:530.13% ETA:701722s Tx:2998 TPS:0.01 C:149.95K NMB:3.66 Pool:1 IO:5998 Network Time:11000.0s
B:4000/10000 40.0% ABT:102.48s Diff:4.217M H:10000 Infl:459.67% ETA:614895s Tx:3998 TPS:0.01 C:199.95K NMB:4.88 Pool:1 IO:7998 Network Time:11000.0s
B:5000/10000 50.0% ABT:102.31s Diff:4.18M H:10000 Infl:431.75% ETA:511535s Tx:4998 TPS:0.01 C:249.95K NMB:6.1 Pool:1 IO:9998 Network Time:11000.0s
B:6000/10000 60.0% ABT:104.77s Diff:4.18M H:10000 Infl:403.57% ETA:419094s Tx:5998 TPS:0.01 C:299.95K NMB:7.32 Pool:1 IO:11998 Network Time:11000.0s
B:7000/10000 70.0% ABT:96.04s Diff:4.036M H:10000 Infl:432.66% ETA:288114s Tx:6998 TPS:0.01 C:349.95K NMB:8.54 Pool:1 IO:13998 Network Time:11000.0s
B:8000/10000 80.0% ABT:101.16s Diff:4.036M H:10000 Infl:399.77% ETA:202314s Tx:7998 TPS:0.01 C:399.95K NMB:9.76 Pool:1 IO:15998 Network Time:11000.0s
B:9000/10000 90.0% ABT:101.7s Diff:4.102M H:10000 Infl:391.17% ETA:101698s Tx:8998 TPS:0.01 C:449.95K NMB:10.98 Pool:1 IO:17998 Network Time:11000.0s
End: B:10000/10000 100.0% ABT:103.16s Diff:4.102M H:10000 Infl:380.2% ETA:0s Tx:9998 TPS:0.01 C:499.95K NMB:12.21 Pool:1 IO:19998 Network Time:11000.0s
Avg Broadcast Time per block: 5.49935
Total Broadcast Time: 109987.0
```

It can be seen that the first few average blocktimes were clearly greater than the target time of 100, but the difficulty was adjusted considerably, to achieve a more reasonable blocktime. This of course could change with a change in transaction frequency as the blocks grow in size with transactions strictly dependent on bandwidth. Latency is not affected by transaction times.

How this works with bandwidth and transactions can be shown below:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 100 --blocksize 4000 --wallets 100 --transactions 100 --interval 10 --print 10 --reward 50 --halving 0 --years 10 --bandwidth 256 --latency 1 --bl
ocks 200
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 100 | Hashrate: 10000 | Blocktime: 100.0 | Print: 10 | Transactions: 100 | Blocksize: 4000 | Interval: 10.0 | R
eward: 50.0 | Halving: 0 | Years: 10 | Blocks: 200 | Difficulty: None | Latency: 1.0 | Bandwidth: 256 | Fee: 0
B:10/200 5.0% ABT:230.86s Diff:5.0M H:10000 Infl:0% ETA:43863s Tx:414 TPS:0.3 C:0.45K NMB:0.11 Pool:1 IO:18 Network Time:909.0s
B:20/200 10.0% ABT:757.93s Diff:5.0M H:10000 Infl:20430.91% ETA:136427s Tx:2530 TPS:0.65 C:0.95K NMB:0.64 Pool:1 IO:38 Network Time:4322.0s
B:30/200 15.0% ABT:1345.9s Diff:5.0M H:10000 Infl:8627.7% ETA:228802s Tx:6425 TPS:0.7 C:1.45K NMB:1.6 Pool:1 IO:58 Network Time:7880.0s
B:40/200 20.0% ABT:880.66s Diff:5.0M H:10000 Infl:12100.38% ETA:148905s Tx:9032 TPS:0.74 C:1.95K NMB:2.24 Pool:1 IO:78 Network Time:5304.0s
B:50/200 25.0% ABT:491.41s Diff:5.0M H:10000 Infl:14210.98% ETA:73712s Tx:10050 TPS:0.37 C:2.45K NMB:2.5 Pool:1 IO:98 Network Time:2126.0s
B:60/200 30.0% ABT:78.89s Diff:5.0M H:10000 Infl:55933.35% ETA:11044s Tx:10060 TPS:0.01 C:2.95K NMB:2.51 Pool:1 IO:118 Network Time:110.0s
B:70/200 35.0% ABT:110.32s Diff:5.0M H:10000 Infl:37135.23% ETA:14341s Tx:10070 TPS:0.01 C:3.45K NMB:2.53 Pool:1 IO:138 Network Time:110.0s
B:80/200 40.0% ABT:104.31s Diff:5.0M H:10000 Infl:38693.27% ETA:12517s Tx:10080 TPS:0.01 C:3.95K NMB:2.54 Pool:1 IO:158 Network Time:110.0s
B:90/200 45.0% ABT:82.27s Diff:5.0M H:10000 Infl:49850.19% ETA:9049s Tx:10090 TPS:0.01 C:4.45K NMB:2.55 Pool:1 IO:178 Network Time:110.0s
B:100/200 50.0% ABT:76.27s Diff:5.0M H:10000 Infl:53742.29% ETA:7627s Tx:10100 TPS:0.02 C:4.95K NMB:2.56 Pool:1 IO:198 Network Time:110.0s
B:110/200 55.0% ABT:92.7s Diff:5.0M H:10000 Infl:42500.48% ETA:8342s Tx:10110 TPS:0.01 C:5.45K NMB:2.57 Pool:1 IO:218 Network Time:110.0s
B:120/200 60.0% ABT:129.08s Diff:5.0M H:10000 Infl:29157.83% ETA:10326s Tx:10120 TPS:0.01 C:5.95K NMB:2.59 Pool:1 IO:238 Network Time:110.0s
B:130/200 65.0% ABT:119.4s Diff:5.0M H:10000 Infl:31538.29% ETA:8357s Tx:10130 TPS:0.01 C:6.45K NMB:2.6 Pool:1 IO:258 Network Time:110.0s
B:140/200 70.0% ABT:161.06s Diff:5.0M H:10000 Infl:22644.08% ETA:9663s Tx:10140 TPS:0.01 C:6.95K NMB:2.61 Pool:1 IO:278 Network Time:110.0s
B:150/200 75.0% ABT:132.03s Diff:5.0M H:10000 Infl:27930.56% ETA:6601s Tx:10150 TPS:0.01 C:7.45K NMB:2.62 Pool:1 IO:298 Network Time:110.0s
B:160/200 80.0% ABT:106.07s Diff:5.0M H:10000 Infl:35397.81% ETA:4242s Tx:10160 TPS:0.01 C:7.95K NMB:2.64 Pool:1 IO:318 Network Time:110.0s
B:170/200 85.0% ABT:143.99s Diff:5.0M H:10000 Infl:25204.28% ETA:4319s Tx:10170 TPS:0.01 C:8.45K NMB:2.65 Pool:1 IO:338 Network Time:110.0s
B:180/200 90.0% ABT:104.96s Diff:5.0M H:10000 Infl:35548.81% ETA:2099s Tx:10180 TPS:0.01 C:8.95K NMB:2.66 Pool:1 IO:358 Network Time:110.0s
B:190/200 95.0% ABT:94.32s Diff:5.0M H:10000 Infl:39965.13% ETA:943s Tx:10190 TPS:0.01 C:9.45K NMB:2.67 Pool:1 IO:378 Network Time:110.0s
End: B:200/200 100.0% ABT:76.72s Diff:5.0M H:10000 Infl:50525.06% ETA:0s Tx:10200 TPS:0.02 C:9.95K NMB:2.68 Pool:1 IO:398 Network Time:110.0s
Avg Broadcast Time per block: 55.4775
Total Broadcast Time: 22191.0
```

It can be seen that the number of transactions begins very high which affects the network time. But network time decreases as the number of transactions decreases.

Fees

Fees are represented in a percentage based fee structure. The input is percent of the transaction. These fees are then given to the winning miner.

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SU25/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 100 --blocksize 4000 --wallets 100 --transactions 100 --interval 10 --print 10 --reward 50 --halving 0 --years 10 --blocks 200 --fee .01
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 100 | Hashrate: 10000 | Blocktime: 100.0 | Print: 10 | Transactions: 100 | Blocksize: 4000 | Interval: 10.0 | R
eward: 50.0 | Halving: 0 | Years: 10 | Blocks: 200 | Difficulty: None | Latency: 0 | Bandwidth: inf | Fee: 0.01
B:10/200 5.0% ABT:37.64s Diff:5.0M H:10000 Infl:0% ETA:7152s Tx:147 TPS:0.39 C:0.45K NMB:0.04 Pool:1 IO:18 AFB:0.06
B:20/200 10.0% ABT:163.02s Diff:5.0M H:10000 Infl:40839.67% ETA:29343s Tx:1298 TPS:0.71 C:0.95K NMB:0.34 Pool:1 IO:38 AFB:0.84
B:30/200 15.0% ABT:143.01s Diff:5.0M H:10000 Infl:33656.89% ETA:24312s Tx:2309 TPS:0.71 C:1.45K NMB:0.59 Pool:1 IO:58 AFB:1.43
B:40/200 20.0% ABT:80.18s Diff:5.0M H:10000 Infl:52895.95% ETA:12828s Tx:2879 TPS:0.71 C:1.95K NMB:0.74 Pool:1 IO:78 AFB:1.14
B:50/200 25.0% ABT:104.07s Diff:5.0M H:10000 Infl:38072.64% ETA:15610s Tx:3617 TPS:0.71 C:2.45K NMB:0.93 Pool:1 IO:98 AFB:1.92
B:60/200 30.0% ABT:83.58s Diff:5.0M H:10000 Infl:45434.25% ETA:11700s Tx:4215 TPS:0.72 C:2.95K NMB:1.09 Pool:1 IO:118 AFB:1.99
B:70/200 35.0% ABT:81.85s Diff:5.0M H:10000 Infl:45058.47% ETA:10640s Tx:4799 TPS:0.71 C:3.45K NMB:1.24 Pool:1 IO:138 AFB:2.38
B:80/200 40.0% ABT:164.05s Diff:5.0M H:10000 Infl:22009.37% ETA:19686s Tx:5957 TPS:0.71 C:3.95K NMB:1.53 Pool:1 IO:158 AFB:5.14
B:90/200 45.0% ABT:112.98s Diff:5.0M H:10000 Infl:31446.38% ETA:12427s Tx:6758 TPS:0.71 C:4.45K NMB:1.74 Pool:1 IO:178 AFB:4.16
B:100/200 50.0% ABT:55.45s Diff:5.0M H:10000 Infl:63258.38% ETA:5545s Tx:7153 TPS:0.71 C:4.95K NMB:1.84 Pool:1 IO:198 AFB:2.28
B:110/200 55.0% ABT:96.58s Diff:5.0M H:10000 Infl:35951.54% ETA:8692s Tx:7842 TPS:0.71 C:5.45K NMB:2.02 Pool:1 IO:218 AFB:4.36
B:120/200 60.0% ABT:77.23s Diff:5.0M H:10000 Infl:44578.36% ETA:6178s Tx:8391 TPS:0.71 C:5.95K NMB:2.16 Pool:1 IO:238 AFB:3.93
B:130/200 65.0% ABT:144.42s Diff:5.0M H:10000 Infl:23670.74% ETA:10109s Tx:9416 TPS:0.71 C:6.45K NMB:2.42 Pool:1 IO:258 AFB:7.73
B:140/200 70.0% ABT:43.08s Diff:5.0M H:10000 Infl:78877.02% ETA:2584s Tx:9727 TPS:0.72 C:6.95K NMB:2.51 Pool:1 IO:278 AFB:2.59
B:150/200 75.0% ABT:119.91s Diff:5.0M H:10000 Infl:28192.1% ETA:5995s Tx:10150 TPS:0.35 C:7.45K NMB:2.62 Pool:1 IO:298 AFB:3.65
B:160/200 80.0% ABT:62.99s Diff:5.0M H:10000 Infl:53421.08% ETA:2519s Tx:10160 TPS:0.02 C:7.95K NMB:2.64 Pool:1 IO:318 AFB:0.0
B:170/200 85.0% ABT:65.12s Diff:5.0M H:10000 Infl:51476.59% ETA:1953s Tx:10170 TPS:0.02 C:8.45K NMB:2.65 Pool:1 IO:338 AFB:0.0
B:180/200 90.0% ABT:93.61s Diff:5.0M H:10000 Infl:35682.89% ETA:1872s Tx:10180 TPS:0.01 C:8.95K NMB:2.66 Pool:1 IO:358 AFB:0.0
B:190/200 95.0% ABT:172.75s Diff:5.0M H:10000 Infl:19275.46% ETA:1727s Tx:10190 TPS:0.01 C:9.45K NMB:2.67 Pool:1 IO:378 AFB:0.0
End: B:200/200 100.0% ABT:89.64s Diff:5.0M H:10000 Infl:37042.93% ETA:0s Tx:10200 TPS:0.01 C:9.95K NMB:2.68 Pool:1 IO:398 AFB:0.0
Total Fees: 435.8996879595618
```

The fees are shown with the label AFB which represents average fees per block. This is the total coins that are taken by fees. Total Fees are also shown at the bottom which are the total fees collected in the simulation.

These fees are created by using wallets with balances and transactions that are logical and possible. Specifically the only transactions can start with the miners, but are given to other wallets throughout.

These fees as well are not applied to the reward transactions.

Latency, Bandwidth, and Fees with Various Coins

Below are the coins with the medium amount of transactions with a bandwidth = 1MB/s, Latency = .1s, fee = 1%.

BTC Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/litt/SU25/CSS95/project_2 (tx-change??) $ python3 sim-blockchain.py --miners 15 --hashrate 10000 --nodes 2 --neighbors 1 --blocktime 600 --blocksize 4000 --wallets 1000 --transactions 1000 --interval 1 --print 25 --reward 50 --halving 210000 --years 10 --latency .1 --bandwidth 1048576 --fee .01
Miners: 15 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 600.0 | Print: 25 | Transactions: 1000 | Blocksize: 4000 | Interval: 1.0 | Reward: 50.0 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0.1 | Bandwidth: 1048576 | Fee: 0.01
B:25/525600 0.0% ABT:492.69s Diff:90.0M H:10000 Infl:0% ETA:258942948s Tx:69689 TPS:5.67 C:1.22K NMB:17.04 Pool:30697 IO:48 Network Time:36.47s AFB:0.92
B:50/525600 0.01% ABT:517.65s Diff:90.0M H:10000 Infl:5001.22% ETA:272048630s Tx:169714 TPS:7.76 C:2.5K NMB:41.48 Pool:131584 IO:98 Network Time:51.39s AFB:1.08
B:75/525600 0.01% ABT:639.87s Diff:90.0M H:10000 Infl:2979.1% ETA:336265175s Tx:269739 TPS:6.27 C:3.77K NMB:65.93 Pool:302649 IO:148 Network Time:51.39s AFB:0.63
B:100/525600 0.02% ABT:499.08s Diff:90.0M H:10000 Infl:3391.8% ETA:262265481s Tx:369764 TPS:8.05 C:5.03K NMB:90.37 Pool:413891 IO:198 Network Time:51.39s AFB:0.67
B:125/525600 0.02% ABT:790.42s Diff:90.0M H:10000 Infl:1999.65% ETA:415346524s Tx:469789 TPS:5.08 C:6.29K NMB:114.82 Pool:530335 IO:248 Network Time:51.39s AFB:0.27
B:150/525600 0.03% ABT:693.81s Diff:90.0M H:10000 Infl:2188.71% ETA:364563862s Tx:569814 TPS:5.78 C:7.55K NMB:139.26 Pool:430335 IO:298 Network Time:51.39s AFB:0.38
B:175/525600 0.03% ABT:574.73s Diff:90.0M H:10000 Infl:2572.34% ETA:301980023s Tx:669839 TPS:6.99 C:8.82K NMB:163.7 Pool:330335 IO:348 Network Time:51.39s AFB:0.67
B:200/525600 0.04% ABT:675.11s Diff:90.0M H:10000 Infl:2142.59% ETA:354705100s Tx:769864 TPS:5.94 C:10.08K NMB:188.15 Pool:230335 IO:398 Network Time:51.39s AFB:0.5
B:225/525600 0.04% ABT:628.16s Diff:90.0M H:10000 Infl:2266.07% ETA:330017240s Tx:869889 TPS:6.39 C:11.34K NMB:212.59 Pool:130335 IO:448 Network Time:51.39s AFB:0.29
B:250/525600 0.05% ABT:681.48s Diff:90.0M H:10000 Infl:2062.11% ETA:358015781s Tx:969914 TPS:5.89 C:12.59K NMB:237.04 Pool:30335 IO:498 Network Time:51.39s AFB:0.18
End: B:258/525600 0.05% ABT:627.07s Diff:90.0M H:10000 Infl:12350.06% ETA:329423592s Tx:1000256 TPS:11.52 C:12.99K NMB:244.45 Pool:1 IO:514 Network Time:50.58s AFB:0.11
Avg Broadcast Time per block: 0.9973027192344962
Total Broadcast Time: 514.608203125
Total Fees: 140.89465883136768
```

BCH Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SUZ5/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 600 --blocksize 128000 --wallets 1000 --transactions 1000 --interval 1 --print 10 --reward 12.5 --halving 210000 --years 10 --latency .1 --bandwidth 1048576 --fee .01
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 600.0 | Print: 10 | Transactions: 1000 | Blocksize: 128000 | Interval: 1.0
| Reward: 12.5 | Halving: 210000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0.1 | Bandwidth: 1048576 | Fee: 0.01
B:10/525600 0.0% ABT:780.13s Diff:30.0M H:10000 InFl:0% ETA:410029340s Tx:30922 TPS:3.97 C:0.11K NMB:7.56 Pool:15 IO:18 Network Time:16.02s AFB:0.2
B:20/525600 0.0% ABT:843.17s Diff:30.0M H:10000 InFl:8091.42% ETA:443154022s Tx:89746 TPS:7.0 C:0.25K NMB:21.93 Pool:15 IO:38 Network Time:29.74s AFB:0.73
B:30/525600 0.01% ABT:511.04s Diff:30.0M H:10000 InFl:9579.3% ETA:268587945s Tx:125407 TPS:7.0 C:0.38K NMB:30.65 Pool:8 IO:58 Network Time:18.43s AFB:0.99
B:40/525600 0.01% ABT:603.07s Diff:30.0M H:10000 InFl:7171.35% ETA:316949815s Tx:167466 TPS:7.0 C:0.52K NMB:40.92 Pool:22 IO:78 Network Time:21.56s AFB:1.49
B:50/525600 0.01% ABT:742.78s Diff:30.0M H:10000 InFl:5457.45% ETA:390369031s Tx:219269 TPS:7.0 C:0.67K NMB:53.58 Pool:36 IO:98 Network Time:26.31s AFB:2.15
B:60/525600 0.01% ABT:798.2s Diff:30.0M H:10000 InFl:4906.43% ETA:419488299s Tx:274985 TPS:7.0 C:0.83K NMB:67.19 Pool:8 IO:118 Network Time:28.22s AFB:3.37
B:70/525600 0.01% ABT:426.41s Diff:30.0M H:10000 InFl:8769.63% ETA:224092233s Tx:304738 TPS:7.0 C:0.98K NMB:74.47 Pool:8 IO:138 Network Time:15.55s AFB:2.5
B:80/525600 0.02% ABT:450.83s Diff:30.0M H:10000 InFl:8135.68% ETA:236921833s Tx:336185 TPS:7.0 C:1.13K NMB:82.15 Pool:15 IO:158 Network Time:16.37s AFB:3.02
B:90/525600 0.02% ABT:374.32s Diff:30.0M H:10000 InFl:9584.71% ETA:196708652s Tx:362312 TPS:7.0 C:1.28K NMB:88.54 Pool:1 IO:178 Network Time:13.78s AFB:2.61
B:100/525600 0.02% ABT:527.21s Diff:30.0M H:10000 InFl:6793.09% ETA:277048755s Tx:399086 TPS:7.0 C:1.45K NMB:97.53 Pool:8 IO:198 Network Time:18.98s AFB:4.38
B:110/525600 0.02% ABT:860.1s Diff:30.0M H:10000 InFl:4168.4% ETA:451973852s Tx:459086 TPS:7.0 C:1.64K NMB:112.19 Pool:15 IO:218 Network Time:30.32s AFB:6.79
B:120/525600 0.02% ABT:679.79s Diff:30.0M H:10000 InFl:5173.13% ETA:357216952s Tx:506528 TPS:7.0 C:1.83K NMB:123.78 Pool:1 IO:238 Network Time:24.18s AFB:5.78
B:130/525600 0.02% ABT:568.75s Diff:30.0M H:10000 InFl:6104.07% ETA:298860459s Tx:546200 TPS:7.0 C:2.0K NMB:133.48 Pool:8 IO:258 Network Time:20.39s AFB:5.21
B:140/525600 0.03% ABT:834.72s Diff:30.0M H:10000 InFl:4178.35% ETA:438613885s Tx:604429 TPS:7.0 C:2.21K NMB:147.7 Pool:15 IO:278 Network Time:29.45s AFB:7.96
B:150/525600 0.03% ABT:766.25s Diff:30.0M H:10000 InFl:4493.63% ETA:402628167s Tx:657891 TPS:7.0 C:2.4K NMB:160.76 Pool:8 IO:298 Network Time:27.12s AFB:6.94
B:160/525600 0.03% ABT:787.19s Diff:30.0M H:10000 InFl:4374.72% ETA:413621066s Tx:712816 TPS:7.0 C:2.62K NMB:174.18 Pool:1 IO:318 Network Time:27.84s AFB:8.69
B:170/525600 0.03% ABT:760.04s Diff:30.0M H:10000 InFl:4521.36% ETA:399350108s Tx:765830 TPS:7.0 C:2.84K NMB:187.14 Pool:15 IO:338 Network Time:26.91s AFB:9.9
5
B:180/525600 0.03% ABT:617.81s Diff:30.0M H:10000 InFl:5502.99% ETA:324609035s Tx:808939 TPS:7.0 C:3.05K NMB:197.67 Pool:8 IO:358 Network Time:22.07s AFB:8.58
B:190/525600 0.04% ABT:1113.17s Diff:30.0M H:10000 InFl:3076.65% ETA:584871838s Tx:886600 TPS:7.0 C:3.3K NMB:216.64 Pool:1 IO:378 Network Time:38.94s AFB:12.5
8
B:200/525600 0.04% ABT:469.26s Diff:30.0M H:10000 InFl:7154.56% ETA:246551539s Tx:919342 TPS:7.0 C:3.5K NMB:224.64 Pool:1 IO:398 Network Time:17.01s AFB:7.56
B:210/525600 0.04% ABT:832.03s Diff:30.0M H:10000 InFl:4081.14% ETA:437140326s Tx:977368 TPS:7.0 C:3.76K NMB:238.82 Pool:22 IO:418 Network Time:29.35s AFB:13.
05
End: B:215/525600 0.04% ABT:867.2s Diff:30.0M H:10000 InFl:9944.88% ETA:455613914s Tx:1000214 TPS:6.98 C:3.88K NMB:244.4 Pool:1 IO:428 Network Time:30.52s AFB
:11.82
Avg Broadcast Time per block: 1.186520257994186
Total Broadcast Time: 510.20371093750003
Total Fees: 1204.4566170832977
```

LTC Medium:

```
jacob@jacob-hp-ubuntu ~/Documents/lit/SUZ5/CS595/project_2 (tx-change*) $ conda activate lit
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SUZ5/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 150 --blocksize 4000 --wallets 1000 --transactions 1000 --interval 1 --print 25 --reward 50 --halving 810000 --years 10 --latency .1 --bandwidth
1048576 --fee .01
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 150.0 | Print: 25 | Transactions: 1000 | Blocksize: 4000 | Interval: 1.0 |
Reward: 50.0 | Halving: 810000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0.1 | Bandwidth: 1048576 | Fee: 0.01
B:25/2102400 0.0% ABT:90.42s Diff:7.5M H:10000 InFl:0% ETA:190098467s Tx:12412 TPS:5.51 C:1.21K NMB:3.05 Pool:8 IO:48 Network Time:8.51s AFB:0.41
5
End: B:909/2102400 0.04% ABT:154.37s Diff:7.5M H:10000 InFl:15396.93% ETA:324402551s Tx:1000912 TPS:6.84 C:73.06K NMB:245.25 Pool:1 IO:1816 Network Time:15.54
s AFB:86.07
Avg Broadcast Time per block: 0.3197469746974697
Total Broadcast Time: 581.3
Total Fees: 27657.644614315857
```

Dogecoin Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SUZ5/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 60 --blocksize 4000 --wallets 1000 --transactions 1000 --interval 1 --print 250 --reward 10000 --halving 0 --years 10 --latency .1 --bandwidth 10
48576 --fee .01
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 60.0 | Print: 250 | Transactions: 1000 | Blocksize: 4000 | Interval: 1.0 |
Reward: 10000.0 | Halving: 0 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0.1 | Bandwidth: 1048576 | Fee: 0.01
B:250/5256000 0.0% ABT:54.97s Diff:3.0M H:10000 InFl:0% ETA:288884747s Tx:94100 TPS:6.88 C:2490.0K NMB:23.22 Pool:8 IO:498 Network Time:71.33s AFB:612.32
B:500/5256000 0.01% ABT:65.4s Diff:3.0M H:10000 InFl:3884.9% ETA:343688270s Tx:208226 TPS:7.02 C:4990.0K NMB:51.32 Pool:1 IO:998 Network Time:81.21s AFB:2050.
85
B:750/5256000 0.01% ABT:58.87s Diff:3.0M H:10000 InFl:3232.63% ETA:309401887s Tx:310977 TPS:7.02 C:7490.0K NMB:76.65 Pool:1 IO:1498 Network Time:75.66s AFB:28
94.12
B:1000/5256000 0.02% ABT:66.63s Diff:3.0M H:10000 InFl:2537.66% ETA:350138000s Tx:427252 TPS:7.02 C:9990.0K NMB:105.29 Pool:1 IO:1998 Network Time:82.26s AFB:
4174.9
B:1250/5256000 0.02% ABT:62.68s Diff:3.0M H:10000 InFl:2528.79% ETA:329378383s Tx:536646 TPS:7.02 C:12490.0K NMB:132.24 Pool:1 IO:2498 Network Time:78.9s AFB:
4826.46
B:1500/5256000 0.03% ABT:58.46s Diff:3.0M H:10000 InFl:2603.23% ETA:307162020s Tx:638669 TPS:7.02 C:14990.0K NMB:157.39 Pool:1 IO:2998 Network Time:75.3s AFB:
5179.93
B:1750/5256000 0.03% ABT:58.28s Diff:3.0M H:10000 InFl:2538.63% ETA:306205250s Tx:740377 TPS:7.02 C:17490.0K NMB:182.46 Pool:1 IO:3498 Network Time:75.15s AFB:
5806.84
B:2000/5256000 0.04% ABT:53.5s Diff:3.0M H:10000 InFl:2709.25% ETA:281089505s Tx:833755 TPS:7.02 C:19990.0K NMB:205.51 Pool:1 IO:3998 Network Time:71.08s AFB:
5902.76
B:2250/5256000 0.04% ABT:63.46s Diff:3.006M H:10000 InFl:2247.55% ETA:333416541s Tx:944507 TPS:7.02 C:22490.0K NMB:232.79 Pool:1 IO:4498 Network Time:79.57s A
FB:7440.86
End: B:2371/5256000 0.05% ABT:66.01s Diff:3.006M H:10000 InFl:4011.56% ETA:346801309s Tx:1002371 TPS:6.98 C:23700.0K NMB:247.03 Pool:1 IO:4740 Network Time:81
.62s AFB:8176.93
Avg Broadcast Time per block: 0.15416867800374315
Total Broadcast Time: 731.06787109375
Total Fees: 10711668.435627606
```

Memo Medium:

```
(litt) jacob@jacob-hp-ubuntu ~/Documents/lit/SUZ5/CS595/project_2 (tx-change*) $ python3 sim-blockchain.py --miners 5 --hashrate 10000 --nodes 2 --neighbors 1
--blocktime 3.27 --blocksize 32000 --wallets 1000 --transactions 1000 --interval 1 --print 1000 --reward 51.8457072 --halving 96440000 --years 10 --latency .
1 --bandwidth 1048576 --fee .01
Miners: 5 | Nodes: 2 | Neighbors: 1 | Wallets: 1000 | Hashrate: 10000 | Blocktime: 3.27 | Print: 1000 | Transactions: 1000 | Blocksize: 32000 | Interval: 1.0
| Reward: 51.8457072 | Halving: 96440000 | Years: 10 | Blocks: None | Difficulty: None | Latency: 0.1 | Bandwidth: 1048576 | Fee: 0.01
B:1000/96440367 0.0% ABT:3.52s Diff:0.164M H:10000 InFl:0% ETA:339228645s Tx:24660 TPS:7.25 C:52.69K NMB:7.0 Pool:1 IO:1998 Network Time:113.89s AFB:0.9
B:2000/96440367 0.0% ABT:3.54s Diff:0.164M H:10000 InFl:18731.14% ETA:341256910s Tx:49628 TPS:7.29 C:107.17K NMB:14.07 Pool:8 IO:3998 Network Time:114.14s AFB
:2.63
```

```

B:44000/96440367 0.05% ABT:3.29s Diff:0.15M H:10000 InFl:10245.47% ETA:317420193s Tx:1020675 TPS:7.32 C:3863.43K NMB:292.16 Pool:1 IO:87998 Network Time:113.3
1s AFB:71.5
B:45000/96440367 0.05% ABT:3.17s Diff:0.152M H:10000 InFl:10634.79% ETA:305773777s Tx:1043088 TPS:7.33 C:3985.66K NMB:298.6 Pool:1 IO:89998 Network Time:112.9
s AFB:70.39
End: B:45085/96440367 0.05% ABT:3.16s Diff:0.152M H:10000 InFl:114523.2% ETA:304536609s Tx:1045084 TPS:7.23 C:3996.42K NMB:299.17 Pool:1 IO:90168 Network Time
:112.84s AFB:74.78
Avg Broadcast Time per block: 0.05663468569784851
Total Broadcast Time: 5106.749609375001
Total Fees: 1659012.6330032458

```

Blockchain	Blocks	Total Broadcast time	Total fees
BTC	258	514	140
BCH	215	510	1204
LTC	909	581	27657
Dogecoin	2371	731s	10711668
Memo	45085	5106s	1659013

Notes

It can be seen that the effect of bandwidth and latency clearly had an effect on not only the simulation time but also the amount of blocks created to complete the transactions. With every single blockchain having an increase in the number of blocks compared to the original medium runs except for BTC.

It specifically creates more issues for blockchains with smaller blocktimes as this affects them greatly.

Overall, the latency, bandwidth and fees worked out very well. And although they were only shown with 2 nodes, this effect is increased as the number of nodes increases.

Future Improvements

Although this simulation was relatively extensive there are significant improvements that can be made. Firstly, the miners have equal hashrates, which may be something that impacts how the simulation actually occurs. In reality, this would not occur and there would be miners that have greater odds than others to win blocks. This may affect how the economy shifts. On this same note, the hash rate is constant. It could be improved by having a dynamic or in general steadily growing hasrate. This would be more consistent with real life blockchains.